Politecnico di Torino

Advanced Machine Learning

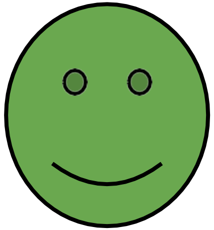
A.A. 2023/2024

Tatiana Tommasi

Domain Adaptation

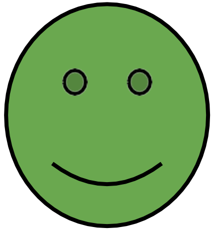
On the success of Deep Learning models...

Is there a bird?



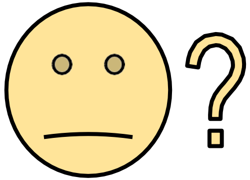
Slide credit: Mancini, Morerio, Tutorial at ICIAP 2019

On the success of Deep Learning models...



Slide credit: Mancini, Morerio, Tutorial at ICIAP 2019

And failure...



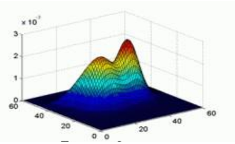
https://bam-dataset.org/

And failure...

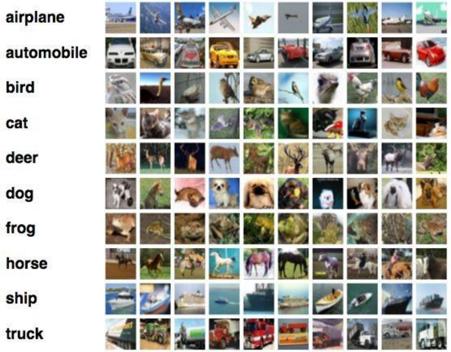
https://bam-dataset.org/

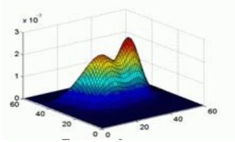
Domain Shift

Visual appearance changes degrade the performance of visual recognition systems.

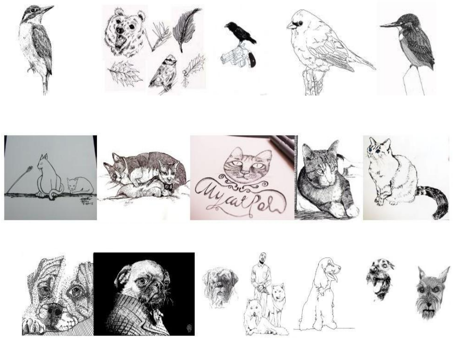


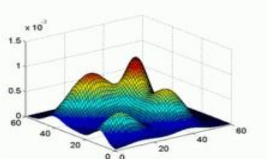
Accuracy: 85%





Accuracy: 55%



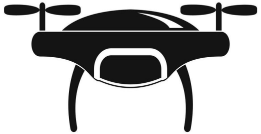


B. Kulis, K. Saenko, and T. Darrell, "What You Saw is Not What You Get: Domain Adaptation Using Asymmetric Kernel Transforms" CVPR, 2011.

Slide credit: Mancini, Morerio, Tutorial at ICIAP 2019

Domain Shift

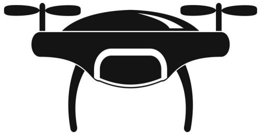
Appearance changes due to seasonal and time changes.

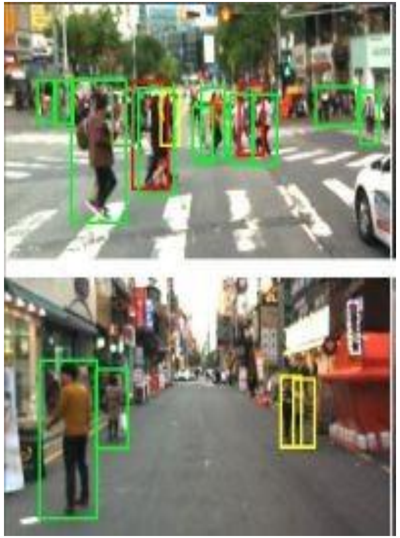


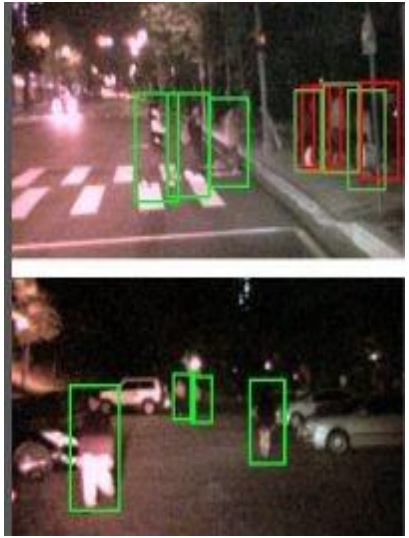
Z. Chen et al., "Deep Learning Features at Scale for Visual Place Recognition". ICRA 2017.

Domain Shift

Appearance changes due to illumination conditions





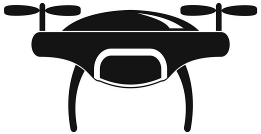


S. Hwang et al., “Multispectral Pedestrian Detection: Benchmark Dataset and Baseline”, CVPR, 2015.

Slide credit: Mancini, Morerio, Tutorial at ICIAP 2019

Domain Shift

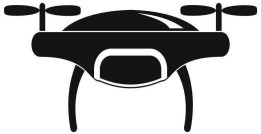
Appearance changes due to different sensors.



S. Hwang et al., “Multispectral Pedestrian Detection: Benchmark Dataset and Baseline”, CVPR, 2015.

Domain Shift

Overcoming costly/unfeasible data collection



https://www.zdnet.com/article/robots-to-the-rescue-searching-for-survivors-checking-on-structural-damage-in-japan/

Slide credit: Mancini, Morerio, Tutorial at ICIAP 2019

Domain Shift

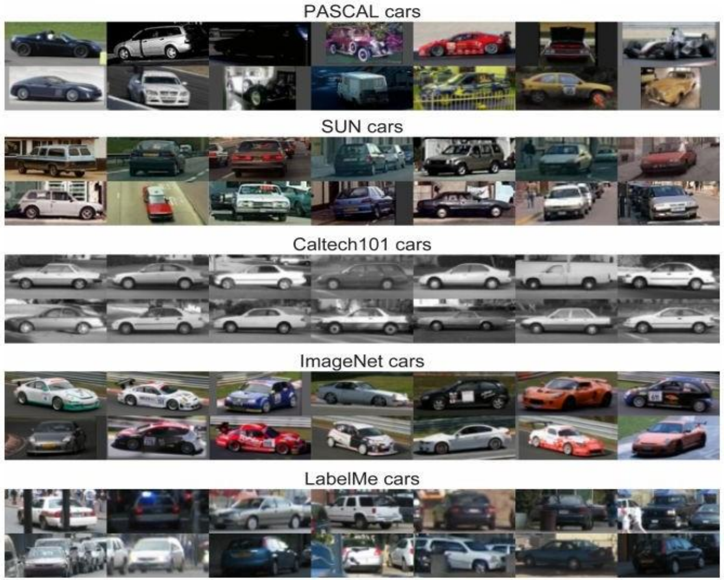
Use of synthetic data

http://synthia-dataset.net/

Slide credit: Mancini, Morerio, Tutorial at ICIAP 2019

Domain Shift and Dataset Bias

Figure 1. Name That Dataset: Given three images from twelve popular object recognition datasets, can you match the images with the dataset? (answer key below)



Torralba and Efros,"Unbiased look at dataset bias, CVPR, 2011.

A bit of Notation

$\mathcal{X}$	Feature space
$\mathcal{Y}$	Output space
$X \in \mathcal{X}$	Input variable
$Y \in \mathcal{Y}$	Output variable
$D = \{\mathcal{X}, P(X)\}$	Domain
$T = \{\mathcal{Y}, P(Y\|X)\}$	Task

The Domain Adaptation Problem

Source Domain

$$D^s = \{\mathcal{X}^s, P(X^s)\}$$
$$T^s = \{\mathcal{Y}^s, P(Y^s | X^s)\}$$

Target Domain

$$D^t = \{\mathcal{X}^t, P(X^t)\}$$
$$T^t = \{\mathcal{Y}^t, P(Y^t | X^t)\}$$

DA problem

$$D^s \neq D^t \qquad T^s = T^t$$

The Domain Adaptation Problem

Source Domain

$$D^s = \{\mathcal{X}^s, P(X^s)\}$$
$$T^s = \{\mathcal{Y}^s, P(Y^s|X^s)\}$$

Target Domain

$$D^t = \{\mathcal{X}^t, P(X^t)\}$$
$$T^t = \{\mathcal{Y}^t, P(Y^t|X^t)\}$$

DA problem

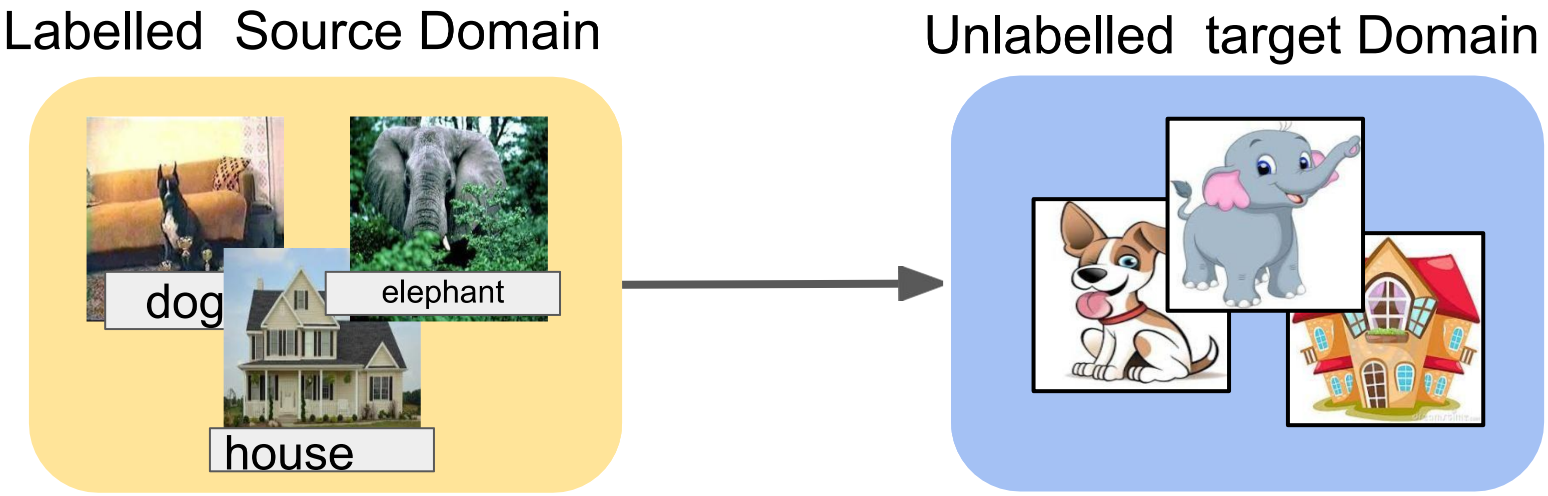
$$\mathcal{X}^s \neq \mathcal{X}^t$$
$$P(X^s) \neq P(X^t)$$

$$\mathcal{Y}^s = \mathcal{Y}^t$$
$$P(Y^s|X^s) = P(Y^t|X^t)$$

covariate shift assumption

The Domain Adaptation Settings

Unsupervised Domain Adaptation

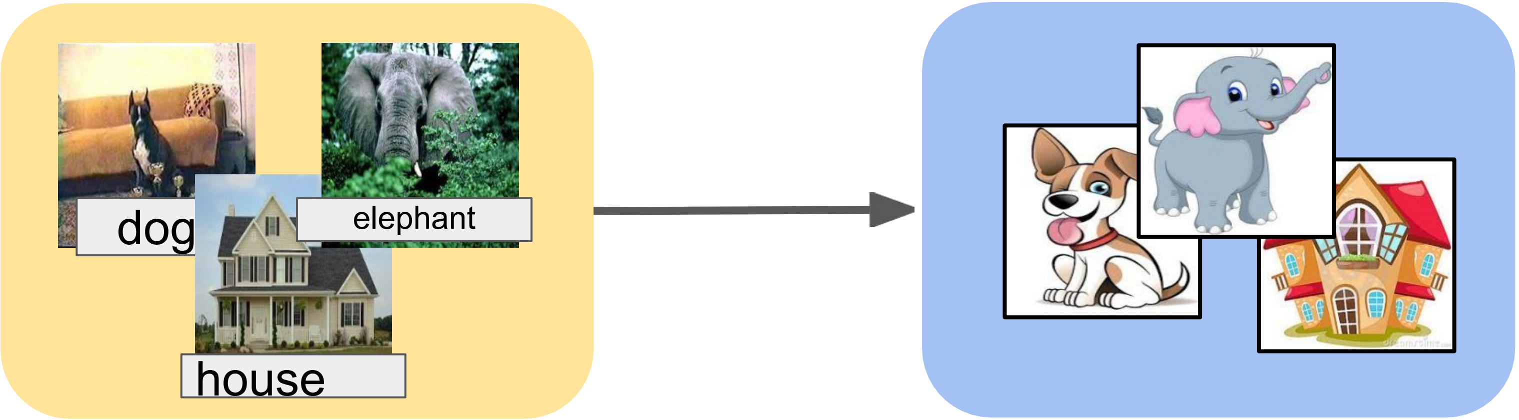


The Domain Adaptation Settings

Unsupervised Domain Adaptation

Labelled Source Domain

Unlabelled target Domain



train

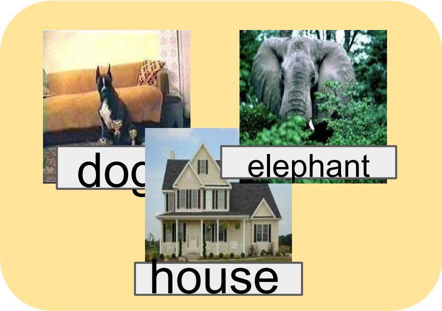
test

Transductive Setting

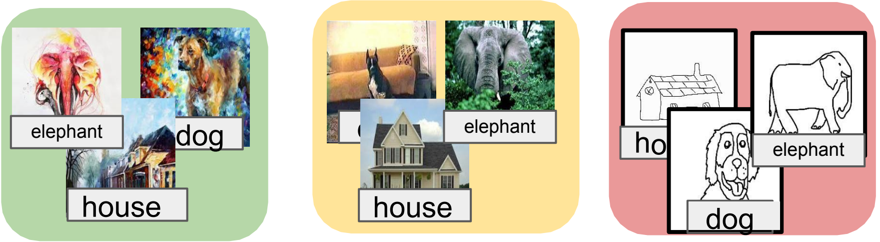
Slide credit: Mancini, Morerio, Tutorial at ICIAP 2019

Variants: number of sources

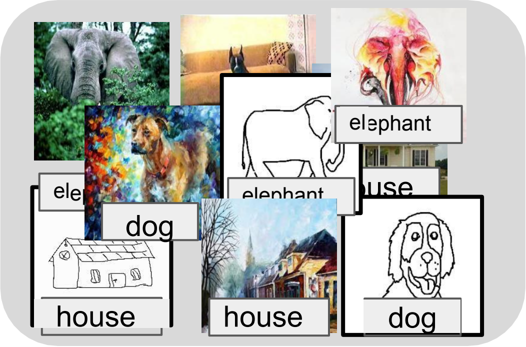
Single



Multi

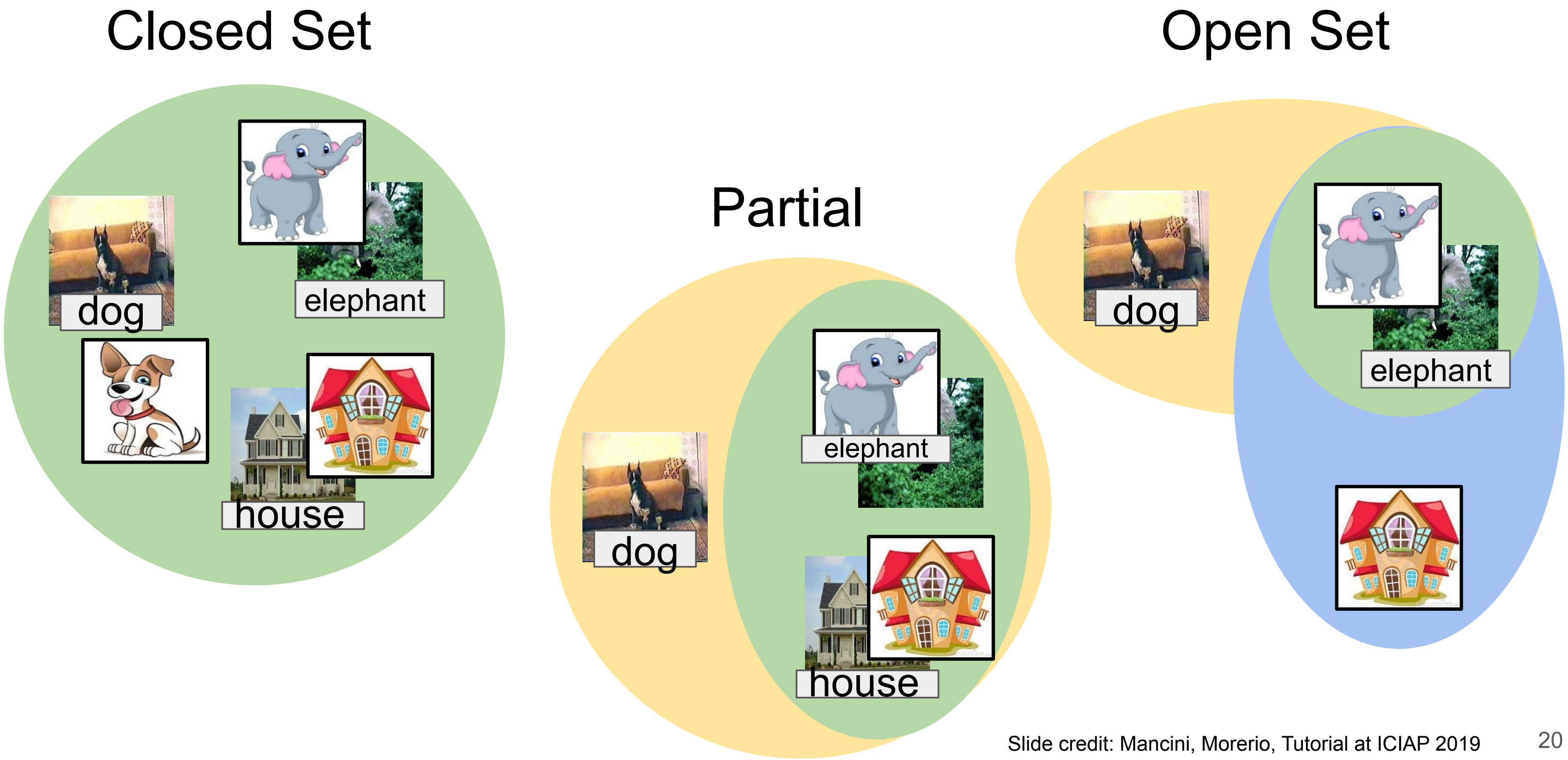


Mixed



Variants: label overlap

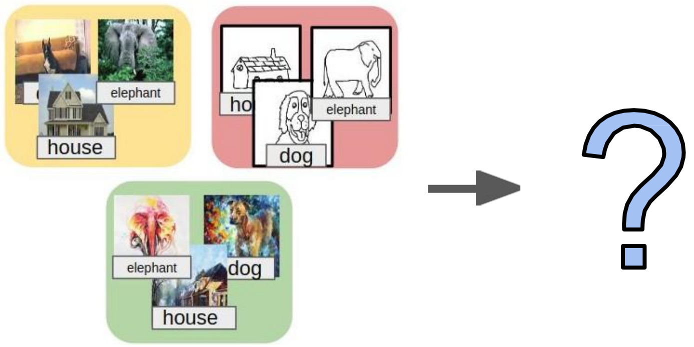
If you want to know more about the DA settings
https://europe.naverlabs.com/eccv-2020-domain-adaptation-tutorial/



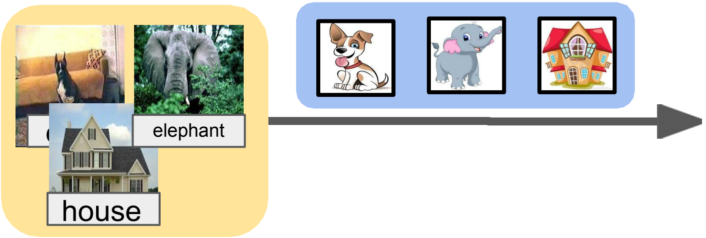
Slide credit: Mancini, Morerio, Tutorial at ICIAP 2019

Without target data!

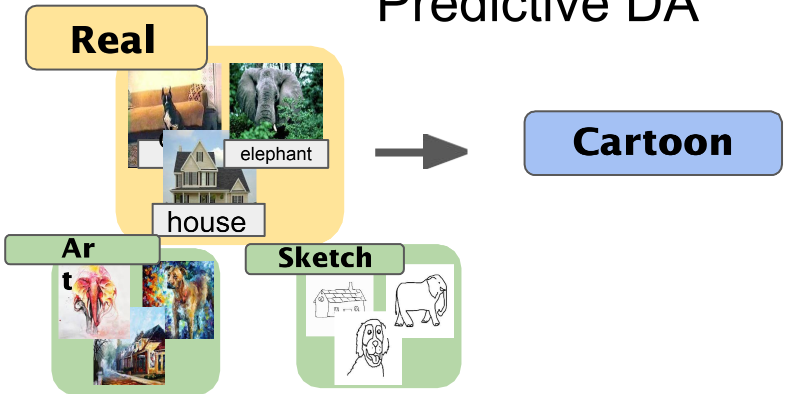
Domain Generalization



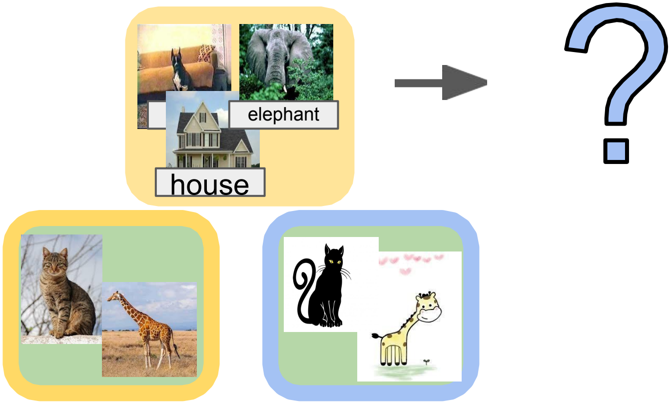
Continuous DA



Predictive DA



Zero-shot DA



Slide credit: Mancini, Morerio, Tutorial at ICIAP 2019

First visual testbed dataset

Office 31

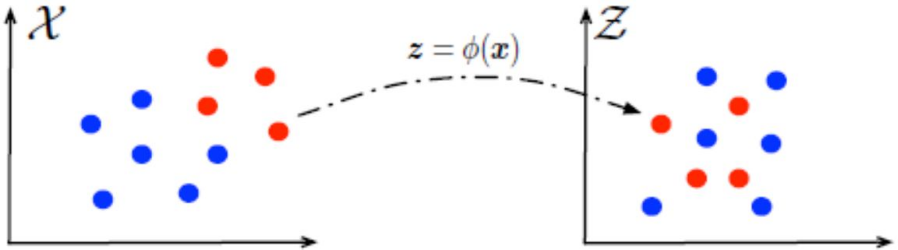
Saenko, K., et al. “Adapting visual category models to new domains.” In ECCV 2010.
Gong, B., et al. “Geodesic flow kernel for unsupervised domain adaptation.” In CVPR 2012.

Slide credit: Mancini, Morerio, Tutorial at ICIAP 2019

First (Shallow) Algorithms

Transfer Component Analysis

Learn a low rank empirical kernel mapping



Mapping implicitly defined by a *kernel* function $K(\cdot,\cdot)$

$$\phi(\boldsymbol{x}_i)^{\mathrm{T}}\phi(\boldsymbol{x}_j) = K(\boldsymbol{x}_i, \boldsymbol{x}_j)$$

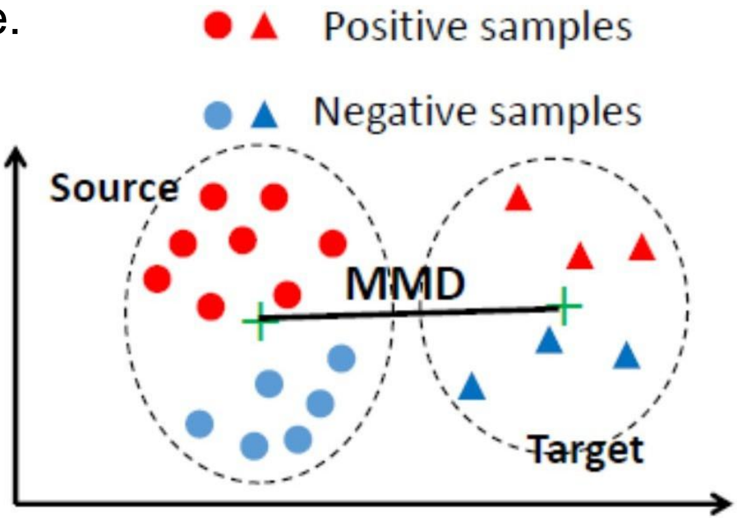
Pan, S., et al. . “Domain Adaptation via Transfer Component Analysis”. In TNN. 2011.
F. Orabona & T. Tommasi. Tutorial on Domain Adaptation and Transfer Learning. In ECCV 2014.

First (Shallow) Algorithms

Maximum Mean Discrepancy

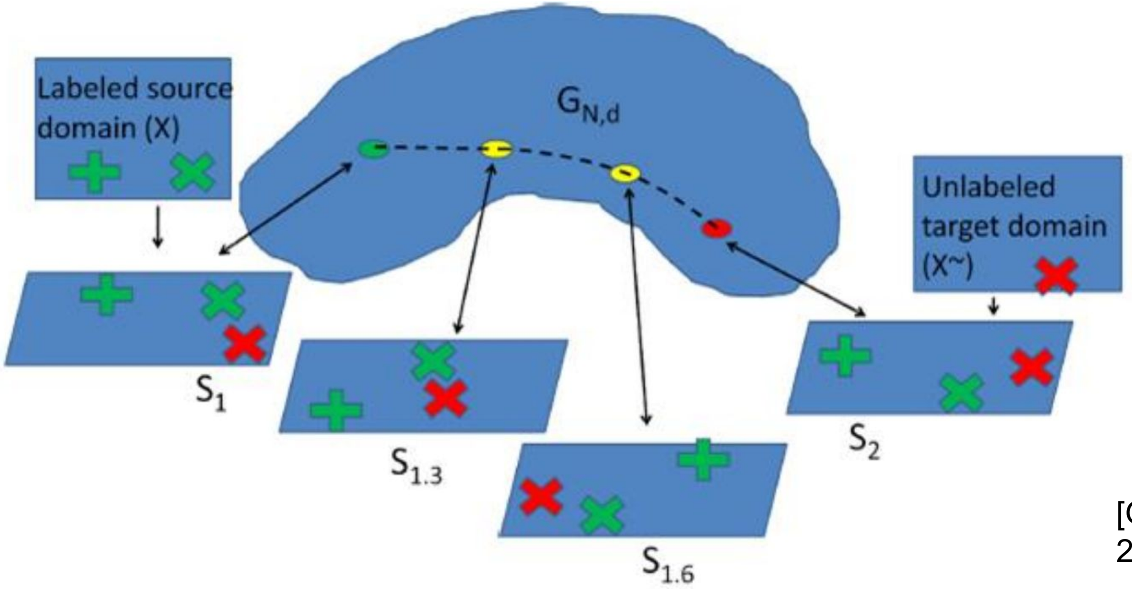
Distance between embeddings of the probability distributions in a reproducing kernel Hilbert space.

$$\mathrm{MMD}^2 = \left\| \frac{1}{n^s} \sum_{i=1}^{n^s} \phi(x_i^s) - \frac{1}{n^t} \sum_{i=1}^{n^t} \phi(x_i^t) \right\|^2$$



Gretton, A., et al. . “A kernel method for the two-sample-problem”. In NIPS. 2007.
F. Orabona & T. Tommasi. Tutorial on Domain Adaptation and Transfer Learning. In ECCV 2014.

First (Shallow) Algorithms



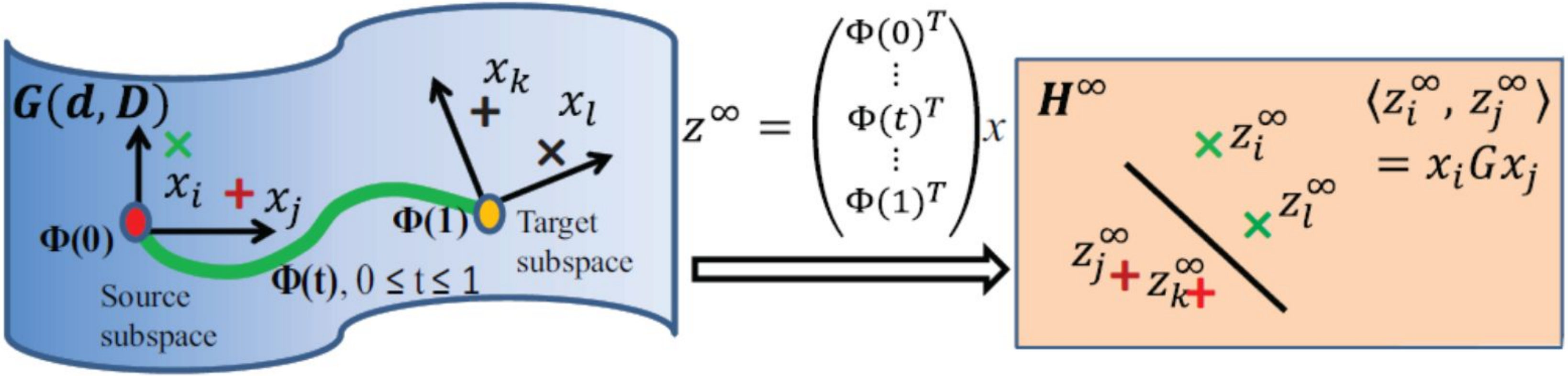
[Gopalan et al. ICCV 2011]

$$\mathbb{G}_{N,d} : d\text{-dimensional vector subspace} \subset \mathbb{R}^N$$

- Apply PCA on source data $\Rightarrow$ matrix S_1 of rank d
- Apply PCA on target data $\Rightarrow$ matrix S_2 of rank d
- Geodesic path on the Grassman manifold between S_1 and S_2

F. Orabona & T. Tommasi. Tutorial on Domain Adaptation and Transfer Learning. In ECCV 2014.

First (Shallow) Algorithms



Geodesic Flow Kernel

[Gong et al. CVPR 2012]

$$\langle \boldsymbol{z}_i^\infty, \boldsymbol{z}_j^\infty \rangle = \int_0^1 (\boldsymbol{\Phi}(t)^{\mathrm{T}} \boldsymbol{x}_i)^{\mathrm{T}} (\boldsymbol{\Phi}(t)^{\mathrm{T}} \boldsymbol{x}_j)\, dt = \boldsymbol{x}_i^{\mathrm{T}} \boldsymbol{G} \boldsymbol{x}_j$$

First (Shallow) Algorithms

Table 1. Accuracy on *Office-31* dataset with standard unsupervised adaptation protocol (Gong et al., 2013).

Method	$A \rightarrow W$	$D \rightarrow W$	$W \rightarrow D$	$A \rightarrow D$	$D \rightarrow A$	$W \rightarrow A$	Average
TCA	21.5 ± 0.0	50.1 ± 0.0	58.4 ± 0.0	11.4 ± 0.0	8.0 ± 0.0	14.6 ± 0.0	27.3
GFK	19.7 ± 0.0	49.7 ± 0.0	63.1 ± 0.0	10.6 ± 0.0	7.9 ± 0.0	15.8 ± 0.0	27.8

Table 2. Accuracy on *Office-10* + *Caltech-10* dataset with standard unsupervised adaptation protocol (Gong et al., 2013).

Method	$A \rightarrow C$	$W \rightarrow C$	$D \rightarrow C$	$C \rightarrow A$	$C \rightarrow W$	$C \rightarrow D$	Average
TCA	42.7 ± 0.0	34.1 ± 0.0	35.4 ± 0.0	54.7 ± 0.0	50.5 ± 0.0	50.3 ± 0.0	44.6
GFK	41.4 ± 0.0	26.4 ± 0.0	36.4 ± 0.0	56.2 ± 0.0	43.7 ± 0.0	42.0 ± 0.0	41.0

Long, M., et al. “Learning Transferable Features with Deep Adaptation Networks”. In ICML 2015..

What happens when moving to deep features?

Table 1. Accuracy on *Office-31* dataset with standard unsupervised adaptation protocol (Gong et al., 2013).

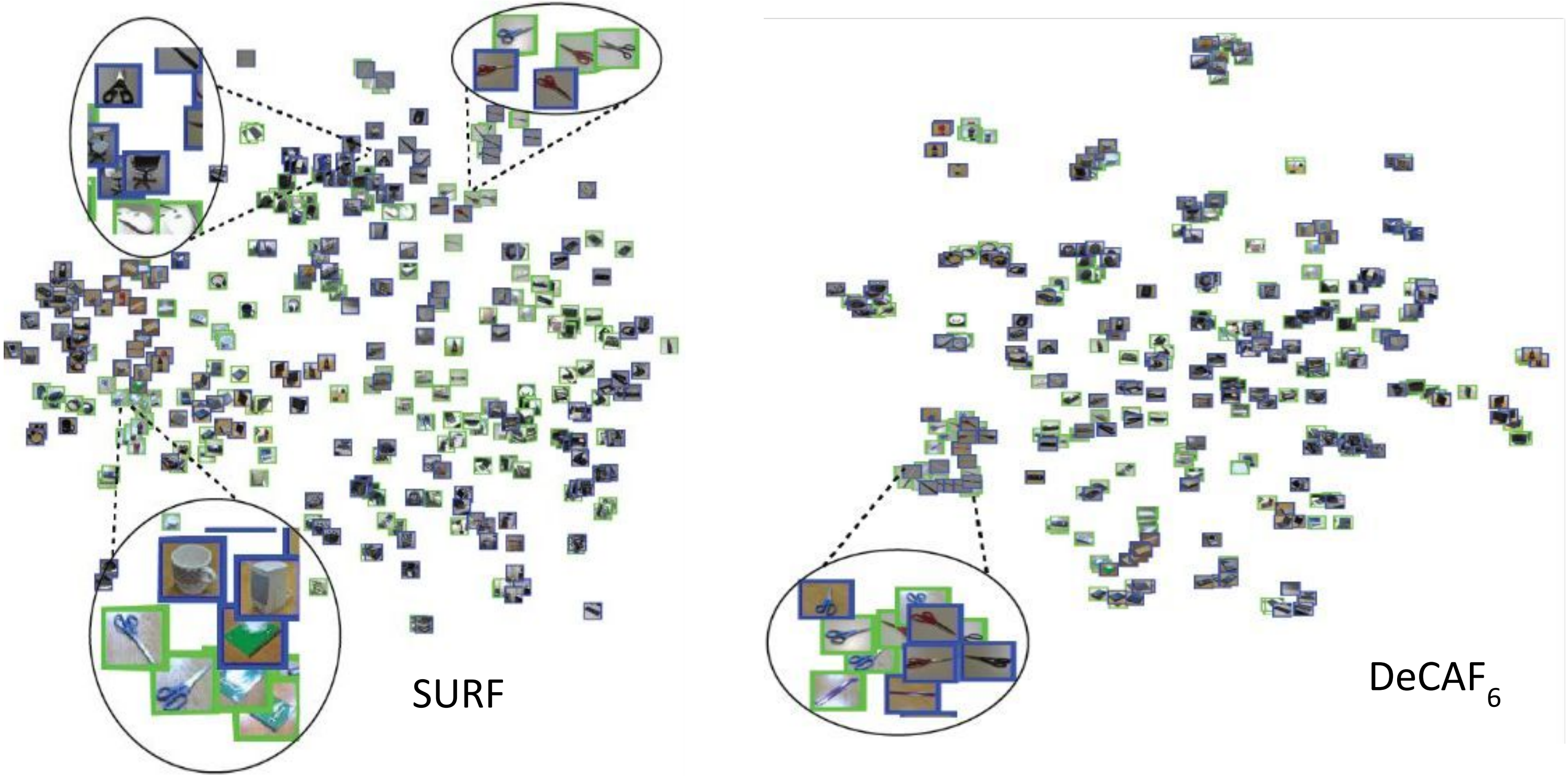
Method	A → W	D → W	W → D	A → D	D → A	W → A	Average
TCA	21.5 ± 0.0	50.1 ± 0.0	58.4 ± 0.0	11.4 ± 0.0	8.0 ± 0.0	14.6 ± 0.0	27.3
GFK	19.7 ± 0.0	49.7 ± 0.0	63.1 ± 0.0	10.6 ± 0.0	7.9 ± 0.0	15.8 ± 0.0	27.8
CNN	61.6 ± 0.5	95.4 ± 0.3	99.0 ± 0.2	63.8 ± 0.5	51.1 ± 0.6	49.8 ± 0.4	70.1

Table 2. Accuracy on *Office-10* + *Caltech-10* dataset with standard unsupervised adaptation protocol (Gong et al., 2013).

Method	A → C	W → C	D → C	C → A	C → W	C → D	Average
TCA	42.7 ± 0.0	34.1 ± 0.0	35.4 ± 0.0	54.7 ± 0.0	50.5 ± 0.0	50.3 ± 0.0	44.6
GFK	41.4 ± 0.0	26.4 ± 0.0	36.4 ± 0.0	56.2 ± 0.0	43.7 ± 0.0	42.0 ± 0.0	41.0
CNN	83.8 ± 0.3	76.1 ± 0.5	80.8 ± 0.4	91.1 ± 0.2	83.1 ± 0.3	89.0 ± 0.3	84.0

Long, M., et al. “Learning Transferable Features with Deep Adaptation Networks”. In ICML 2015..

Deep Features = Universal Representation?



Donahue, J., et al. (2014). Decaf: A deep convolutional activation feature for generic visual recognition. In ICML.

Table 1. Accuracy on *Office-31* dataset with standard unsupervised adaptation protocol (Gong et al., 2013).

Method	A → W	D → W	W → D	A → D	D → A	W → A	Average
TCA	21.5 ± 0.0	50.1 ± 0.0	58.4 ± 0.0	11.4 ± 0.0	8.0 ± 0.0	14.6 ± 0.0	27.3
GFK	19.7 ± 0.0	49.7 ± 0.0	63.1 ± 0.0	10.6 ± 0.0	7.9 ± 0.0	15.8 ± 0.0	27.8
CNN	61.6 ± 0.5	95.4 ± 0.3	<u>99.0</u> ± 0.2	63.8 ± 0.5	51.1 ± 0.6	49.8 ± 0.4	70.1
LapCNN	60.4 ± 0.3	94.7 ± 0.5	**99.1** ± 0.2	63.1 ± 0.6	51.6 ± 0.4	48.2 ± 0.5	69.5
DDC	61.8 ± 0.4	95.0 ± 0.5	98.5 ± 0.4	64.4 ± 0.3	52.1 ± 0.8	<u>52.2</u> ± 0.4	70.6
DAN_7	63.2 ± 0.2	94.8 ± 0.4	98.9 ± 0.3	65.2 ± 0.4	52.3 ± 0.4	52.1 ± 0.4	71.1
DAN_8	<u>63.8</u> ± 0.4	94.6 ± 0.5	98.8 ± 0.6	65.8 ± 0.4	52.8 ± 0.4	51.9 ± 0.5	71.3
DAN_{SK}	63.3 ± 0.3	<u>95.6</u> ± 0.2	<u>99.0</u> ± 0.4	<u>65.9</u> ± 0.7	<u>53.2</u> ± 0.5	52.1 ± 0.4	<u>71.5</u>
DAN	**68.5** ± 0.4	**96.0** ± 0.3	<u>99.0</u> ± 0.2	**67.0** ± 0.4	**54.0** ± 0.4	**53.1** ± 0.3	**72.9**

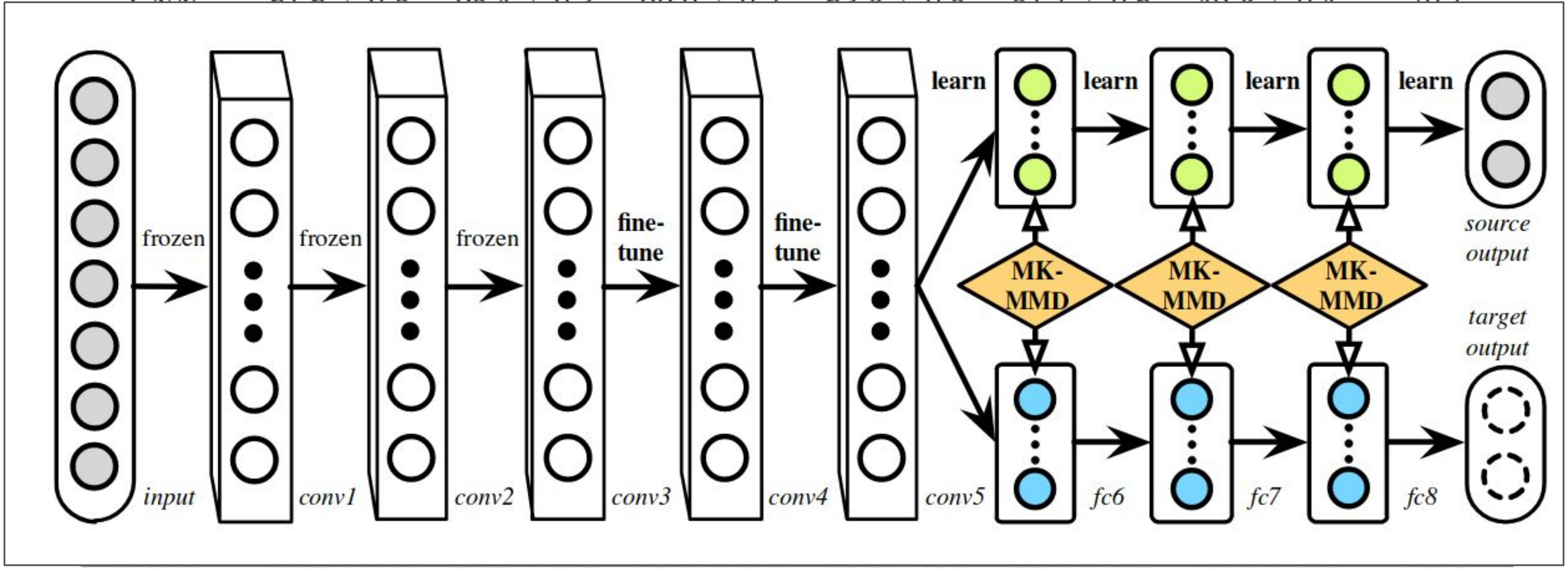
Table 2. Accuracy on *Office-10* + *Caltech-10* dataset with standard unsupervised adaptation protocol (Gong et al., 2013).

Method	A → C	W → C	D → C	C → A	C → W	C → D	Average
TCA	42.7 ± 0.0	34.1 ± 0.0	35.4 ± 0.0	54.7 ± 0.0	50.5 ± 0.0	50.3 ± 0.0	44.6
GFK	41.4 ± 0.0	26.4 ± 0.0	36.4 ± 0.0	56.2 ± 0.0	43.7 ± 0.0	42.0 ± 0.0	41.0
CNN	83.8 ± 0.3	76.1 ± 0.5	80.8 ± 0.4	91.1 ± 0.2	83.1 ± 0.3	89.0 ± 0.3	84.0
LapCNN	83.6 ± 0.6	77.8 ± 0.5	80.6 ± 0.4	**92.1** ± 0.3	81.6 ± 0.4	87.8 ± 0.4	83.9
DDC	84.3 ± 0.5	76.9 ± 0.4	80.5 ± 0.2	91.3 ± 0.3	85.5 ± 0.3	89.1 ± 0.3	84.6
DAN_7	<u>84.7</u> ± 0.3	78.2 ± 0.5	<u>81.8</u> ± 0.3	91.6 ± 0.4	87.4 ± 0.3	88.9 ± 0.5	85.4
DAN_8	84.4 ± 0.3	<u>80.8</u> ± 0.4	81.7 ± 0.2	91.7 ± 0.3	<u>90.5</u> ± 0.4	89.1 ± 0.4	<u>86.4</u>
DAN_{SK}	84.1 ± 0.4	79.9 ± 0.4	81.1 ± 0.5	91.4 ± 0.3	86.9 ± 0.5	<u>89.5</u> ± 0.3	85.5
DAN	**86.0** ± 0.5	**81.5** ± 0.3	**82.0** ± 0.4	<u>92.0</u> ± 0.3	**92.0** ± 0.4	**90.5** ± 0.2	**87.3**

Long, M., et al. “Learning Transferable Features with Deep Adaptation Networks”. In ICML 2015..

Table 1. Accuracy on *Office-31* dataset with standard unsupervised adaptation protocol (Gong et al., 2013).

Method	A → W	D → W	W → D	A → D	D → A	W → A	Average
TCA	21.5 ± 0.0	50.1 ± 0.0	58.4 ± 0.0	11.4 ± 0.0	8.0 ± 0.0	14.6 ± 0.0	27.3
GFK	19.7 ± 0.0	49.7 ± 0.0	63.1 ± 0.0	10.6 ± 0.0	7.9 ± 0.0	15.8 ± 0.0	27.8
DAN_7	84.7 ± 0.3	78.2 ± 0.5	81.8 ± 0.3	91.6 ± 0.4	87.4 ± 0.3	88.9 ± 0.5	85.4
DAN_8	84.4 ± 0.3	80.8 ± 0.4	81.7 ± 0.2	91.7 ± 0.3	90.5 ± 0.4	89.1 ± 0.4	86.4
DAN_{SK}	84.1 ± 0.4	79.9 ± 0.4	81.1 ± 0.5	91.4 ± 0.3	86.9 ± 0.5	89.5 ± 0.3	85.5
DAN	**86.0** ± 0.5	**81.5** ± 0.3	**82.0** ± 0.4	92.0 ± 0.3	**92.0** ± 0.4	**90.5** ± 0.2	**87.3**



Long, M., et al. “Learning Transferable Features with Deep Adaptation Networks”. In ICML 2015..

Office-31 (2017)

Table 1. Classification accuracy (%) on *Office-31* dataset for unsupervised domain adaptation (AlexNet and ResNet)

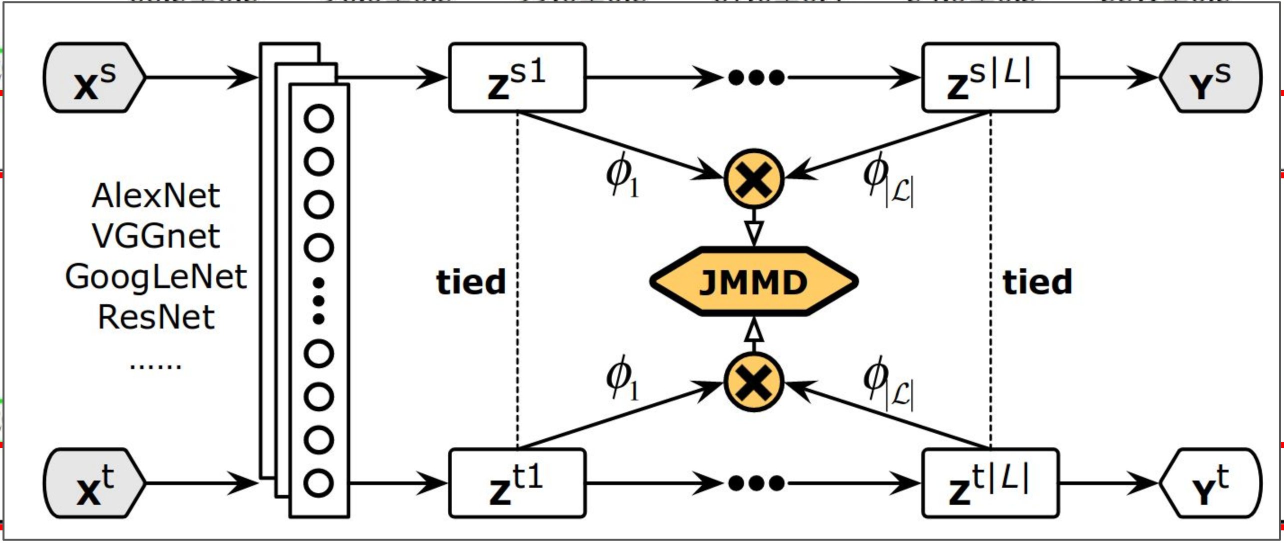
Method	A → W	D → W	W → D	A → D	D → A	W → A	Avg
AlexNet (Krizhevsky et al., 2012)	61.6±0.5	95.4±0.3	99.0±0.2	63.8±0.5	51.1±0.6	49.8±0.4	70.1
TCA (Pan et al., 2011)	61.0±0.0	93.2±0.0	95.2±0.0	60.8±0.0	51.6±0.0	50.9±0.0	68.8
GFK (Gong et al., 2012)	60.4±0.0	95.6±0.0	95.0±0.0	60.6±0.0	52.4±0.0	48.1±0.0	68.7
DDC (Tzeng et al., 2014)	61.8±0.4	95.0±0.5	98.5±0.4	64.4±0.3	52.1±0.6	52.2±0.4	70.6
DAN (Long et al., 2015)	68.5±0.5	96.0±0.3	99.0±0.3	67.0±0.4	54.0±0.5	53.1±0.5	72.9
RTN (Long et al., 2016)	73.3±0.3	**96.8**±0.2	**99.6**±0.1	71.0±0.2	50.5±0.3	51.0±0.1	73.7
RevGrad (Ganin & Lempitsky, 2015)	73.0±0.5	96.4±0.3	99.2±0.3	72.3±0.3	53.4±0.4	51.2±0.5	74.3
JAN (ours)	74.9±0.3	96.6±0.2	99.5±0.2	71.8±0.2	**58.3**±0.3	55.0±0.4	76.0
JAN-A (ours)	**75.2**±0.4	96.6±0.2	**99.6**±0.1	**72.8**±0.3	57.5±0.2	**56.3**±0.2	**76.3**
ResNet (He et al., 2016)	68.4±0.2	96.7±0.1	99.3±0.1	68.9±0.2	62.5±0.3	60.7±0.3	76.1
TCA (Pan et al., 2011)	72.7±0.0	96.7±0.0	99.6±0.0	74.1±0.0	61.7±0.0	60.9±0.0	77.6
GFK (Gong et al., 2012)	72.8±0.0	95.0±0.0	98.2±0.0	74.5±0.0	63.4±0.0	61.0±0.0	77.5
DDC (Tzeng et al., 2014)	75.6±0.2	96.0±0.2	98.2±0.1	76.5±0.3	62.2±0.4	61.5±0.5	78.3
DAN (Long et al., 2015)	80.5±0.4	97.1±0.2	99.6±0.1	78.6±0.2	63.6±0.3	62.8±0.2	80.4
RTN (Long et al., 2016)	84.5±0.2	96.8±0.1	99.4±0.1	77.5±0.3	66.2±0.2	64.8±0.3	81.6
RevGrad (Ganin & Lempitsky, 2015)	82.0±0.4	96.9±0.2	99.1±0.1	79.7±0.4	68.2±0.4	67.4±0.5	82.2
JAN (ours)	85.4±0.3	**97.4**±0.2	**99.8**±0.2	84.7±0.3	68.6±0.3	70.0±0.4	84.3
JAN-A (ours)	**86.0**±0.4	96.7±0.3	99.7±0.1	**85.1**±0.4	**69.2**±0.4	**70.7**±0.5	**84.6**

Long, M., et al. “Deep transfer learning with joint adaptation networks”. In ICML 2017.

Office-31 (2017)

Table 1. Classification accuracy (%) on *Office-31* dataset for unsupervised domain adaptation (AlexNet and ResNet)

Method	A → W	D → W	W → D	A → D	D → A	W → A	Avg
AlexNet (Krizhevsky et al., 2012)	61.6±0.5	95.4±0.3	99.0±0.2	63.8±0.5	51.1±0.6	49.8±0.4	70.1
TCA (Pan et al., 2011)	61.0±0.0	93.2±0.0	95.2±0.0	60.8±0.0	51.6±0.0	50.9±0.0	68.8
GFK (Gong et al., 2012)	60.4±0.0	95.6±0.0	95.0±0.0	60.6±0.0	52.4±0.0	48.1±0.0	68.7
DDC (Tzeng et al., 2014)	61.8±0.4	95.0±0.5	98.5±0.4	64.4±0.3	52.1±0.6	52.2±0.4	70.6
DAN (Long et al., 2015)	68.5±0.5	96.0±0.3	99.0±0.3	67.0±0.4	54.0±0.5	53.1±0.5	72.9
RTN (Long et al., 2016)							73.7
RevGrad (Ganin & Lempitsky, 2							74.3
JAN (ours)							76.0
JAN-A (ours)							**76.3**
ResNet (He et al., 2016)							76.1
TCA (Pan et al., 2011)							77.6
GFK (Gong et al., 2012)							77.5
DDC (Tzeng et al., 2014)							78.3
DAN (Long et al., 2015)							80.4
RTN (Long et al., 2016)							81.6
RevGrad (Ganin & Lempitsky, 2							82.2
JAN (ours)							84.3
JAN-A (ours)							**84.6**



Long, M., et al. “Deep transfer learning with joint adaptation networks”. In ICML 2017.

Office-31 (2017)

Table 1. Classification accuracy (%) on *Office-31* dataset for unsupervised domain adaptation (AlexNet and ResNet)

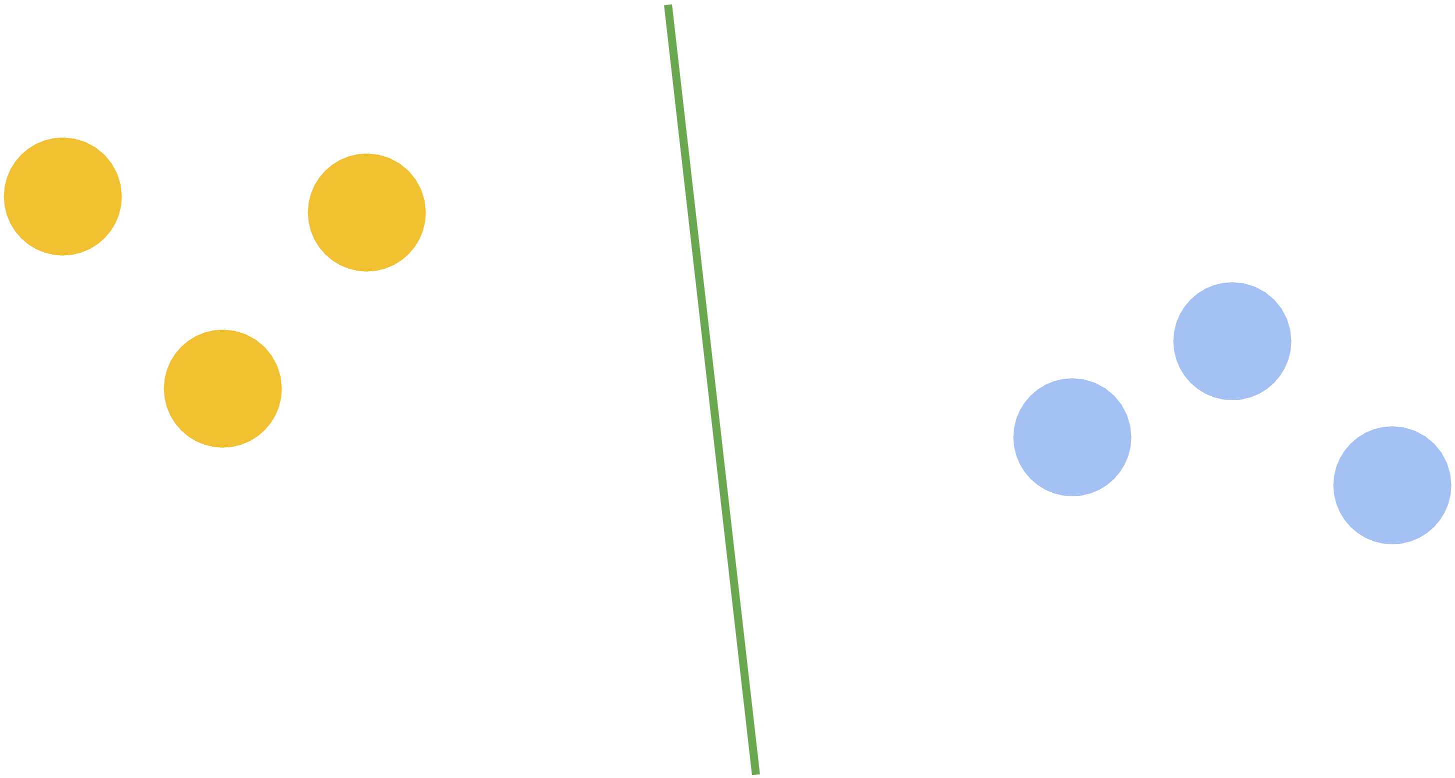
Method	A → W	D → W	W → D	A → D	D → A	W → A	Avg
AlexNet (Krizhevsky et al., 2012)	61.6±0.5	95.4±0.3	99.0±0.2	63.8±0.5	51.1±0.6	49.8±0.4	70.1
TCA (Pan et al., 2011)	61.0±0.0	93.2±0.0	95.2±0.0	60.8±0.0	51.6±0.0	50.9±0.0	68.8
GFK (Gong et al., 2012)	60.4±0.0	95.6±0.0	95.0±0.0	60.6±0.0	52.4±0.0	48.1±0.0	68.7
DDC (Tzeng et al., 2014)	61.8±0.4	95.0±0.5	98.5±0.4	64.4±0.3	52.1±0.6	52.2±0.4	70.6
DAN (Long et al., 2015)	68.5±0.5	96.0±0.3	99.0±0.3	67.0±0.4	54.0±0.5	53.1±0.5	72.9
RTN (Long et al., 2016)	73.3±0.3	**96.8**±0.2	**99.6**±0.1	71.0±0.2	50.5±0.3	51.0±0.1	73.7
RevGrad (Ganin & Lempitsky, 2015)	73.0±0.5	96.4±0.3	99.2±0.3	72.3±0.3	53.4±0.4	51.2±0.5	74.3
JAN (ours)	74.9±0.3	96.6±0.2	99.5±0.2	71.8±0.2	**58.3**±0.3	55.0±0.4	76.0
JAN-A (ours)	**75.2**±0.4	96.6±0.2	**99.6**±0.1	**72.8**±0.3	57.5±0.2	**56.3**±0.2	**76.3**
ResNet (He et al., 2016)	68.4±0.2	96.7±0.1	99.3±0.1	68.9±0.2	62.5±0.3	60.7±0.3	76.1
TCA (Pan et al., 2011)	72.7±0.0	96.7±0.0	99.6±0.0	74.1±0.0	61.7±0.0	60.9±0.0	77.6
GFK (Gong et al., 2012)	72.8±0.0	95.0±0.0	98.2±0.0	74.5±0.0	63.4±0.0	61.0±0.0	77.5
DDC (Tzeng et al., 2014)	75.6±0.2	96.0±0.2	98.2±0.1	76.5±0.3	62.2±0.4	61.5±0.5	78.3
DAN (Long et al., 2015)	80.5±0.4	97.1±0.2	99.6±0.1	78.6±0.2	63.6±0.3	62.8±0.2	80.4
RTN (Long et al., 2016)	84.5±0.2	96.8±0.1	99.4±0.1	77.5±0.3	66.2±0.2	64.8±0.3	81.6
RevGrad (Ganin & Lempitsky, 2015)	82.0±0.4	96.9±0.2	99.1±0.1	79.7±0.4	68.2±0.4	67.4±0.5	82.2
JAN (ours)	85.4±0.3	**97.4**±0.2	**99.8**±0.2	84.7±0.3	68.6±0.3	70.0±0.4	84.3
JAN-A (ours)	**86.0**±0.4	96.7±0.3	99.7±0.1	**85.1**±0.4	**69.2**±0.4	**70.7**±0.5	**84.6**

Adversarial Domain Adaptation (RevGrad)

Slide credit: Mancini, Morerio, Tutorial at ICIAP 2019

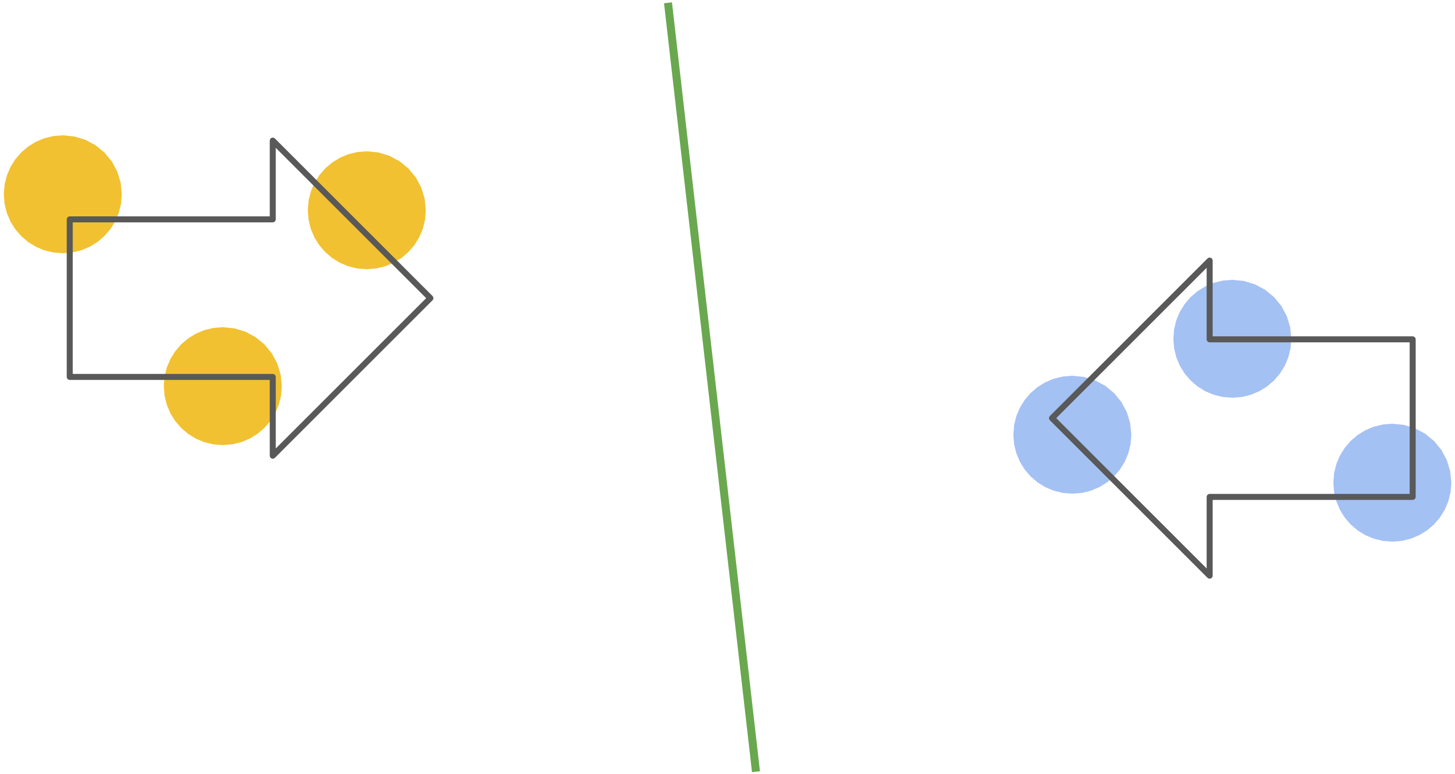
Ganin, Y., et al. “Domain-adversarial training of neural networks”. The Journal of Machine Learning Research, 17(1), 2096-2030, 2016.

Adversarial Domain Adaptation (RevGrad)



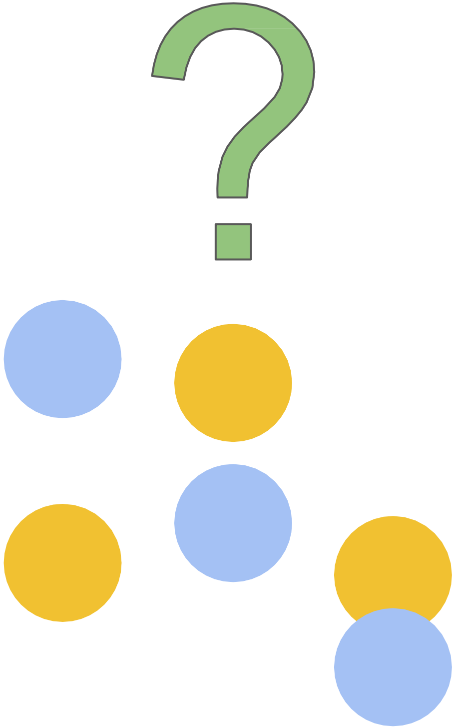
Slide credit: Mancini, Morerio, Tutorial at ICIAP 2019

Adversarial Domain Adaptation (RevGrad)



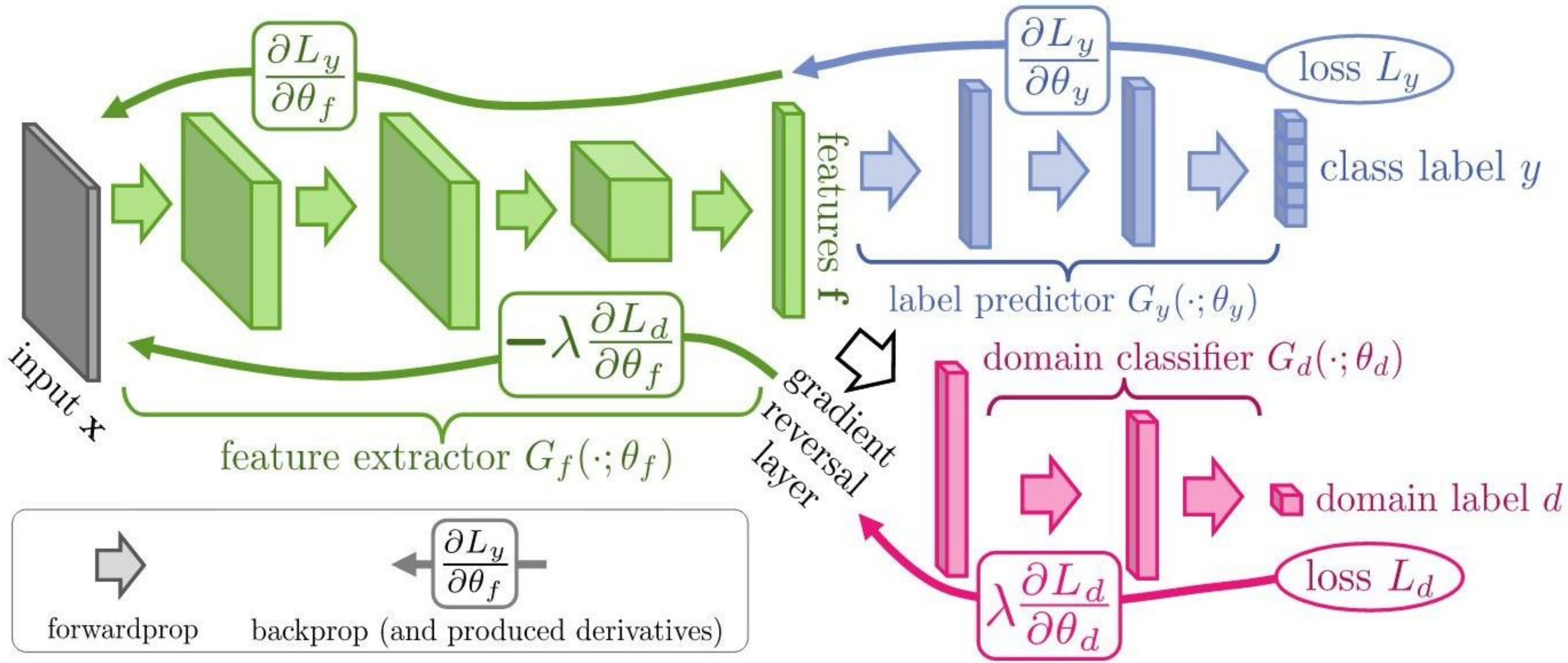
Slide credit: Mancini, Morerio, Tutorial at ICIAP 2019

Adversarial Domain Adaptation (RevGrad)



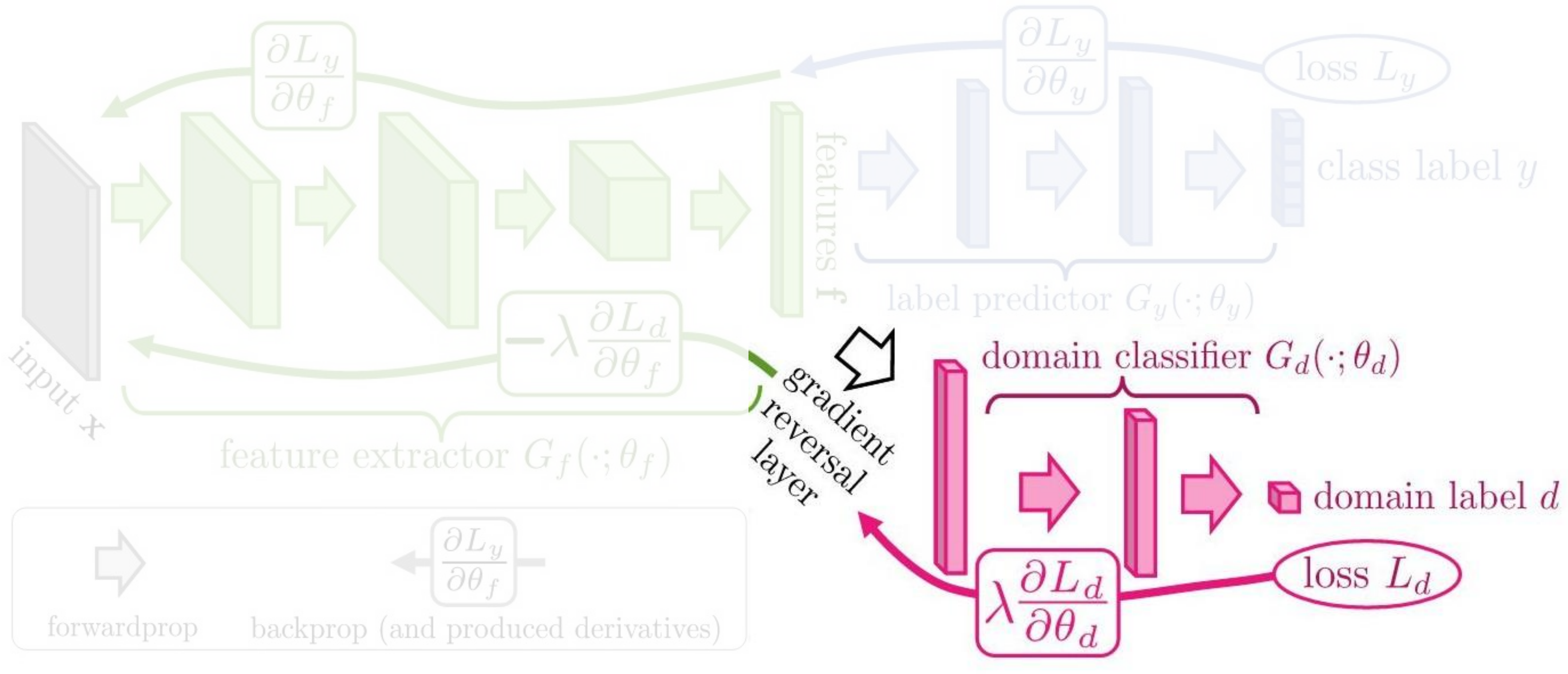
Slide credit: Mancini, Morerio, Tutorial at ICIAP 2019

Adversarial Domain Adaptation (RevGrad-DANN)



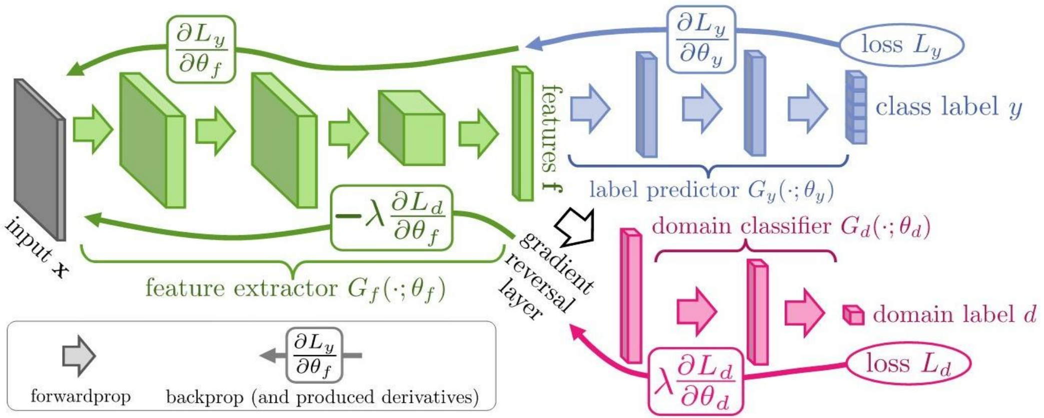
Ganin, Y., et al. "Domain-adversarial training of neural networks". The Journal of Machine Learning Research, 17(1), 2096-2030, 2016.

Adversarial Domain Adaptation (RevGrad-DANN)



Ganin, Y., et al. “Domain-adversarial training of neural networks”. The Journal of Machine Learning Research, 17(1), 2096-2030, 2016.

Adversarial Domain Adaptation (RevGrad-DANN)



$$E(\theta_f,\theta_y,\theta_d) = \sum_{\substack{i=1..N\\d_i=0}} L_y\left(G_y(G_f(\mathbf{x}_i;\theta_f);\theta_y),y_i\right) -$$

$$\lambda \sum_{i=1..N} L_d\left(G_d(G_f(\mathbf{x}_i;\theta_f);\theta_d),y_i\right) =$$

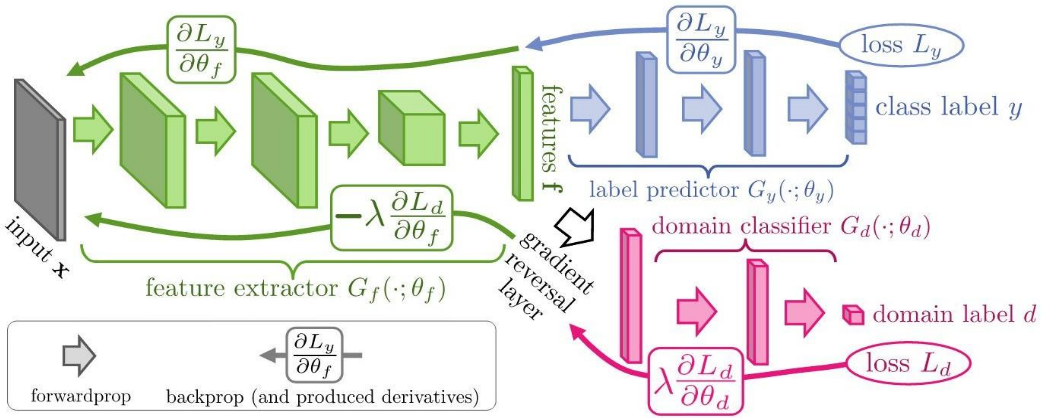
$$= \sum_{\substack{i=1..N\\d_i=0}} L_y^i(\theta_f,\theta_y) - \lambda \sum_{i=1..N} L_d^i(\theta_f,\theta_d) \qquad (1)$$

$$(\hat\theta_f,\hat\theta_y) = \arg\min_{\theta_f,\theta_y} E(\theta_f,\theta_y,\hat\theta_d) \qquad (2)$$

$$\hat\theta_d = \arg\max_{\theta_d} E(\hat\theta_f,\hat\theta_y,\theta_d)\,. \qquad (3)$$

“Unsupervised Domain Adaptation by Backpropagation”. ICML 2015

Adversarial Domain Adaptation (RevGrad-DANN)



We will note the prediction loss and the domain loss respectively by

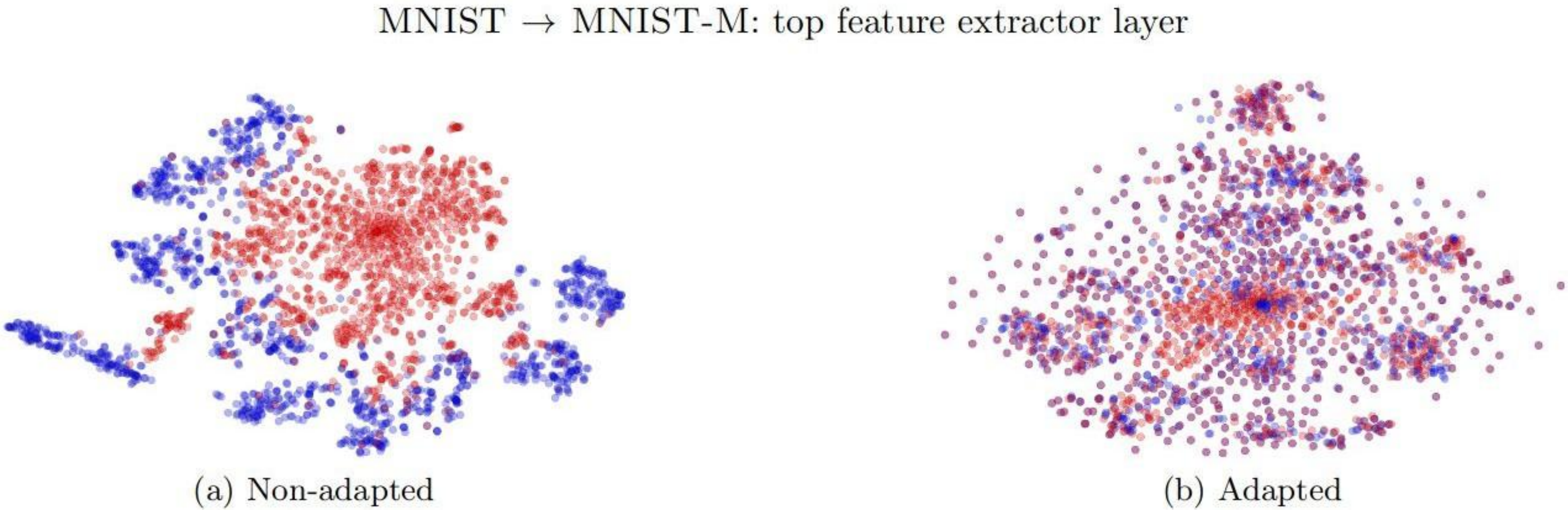
$$\begin{aligned} \mathcal{L}_y^i(\theta_f, \theta_y) &= \mathcal{L}_y\big(G_y(G_f(\mathbf{x}_i;\theta_f);\theta_y), y_i\big)\,, \\ \mathcal{L}_d^i(\theta_f, \theta_d) &= \mathcal{L}_d\big(G_d(G_f(\mathbf{x}_i;\theta_f);\theta_d), d_i\big)\,. \end{aligned}$$

Training DANN then parallels the single layer case and consists in optimizing

$$E(\theta_f, \theta_y, \theta_d) = \frac{1}{n}\sum_{i=1}^{n} \mathcal{L}_y^i(\theta_f, \theta_y) - \lambda\Big(\frac{1}{n}\sum_{i=1}^{n} \mathcal{L}_d^i(\theta_f, \theta_d) + \frac{1}{n'}\sum_{i=n+1}^{N} \mathcal{L}_d^i(\theta_f, \theta_d)\Big), \qquad (10)$$

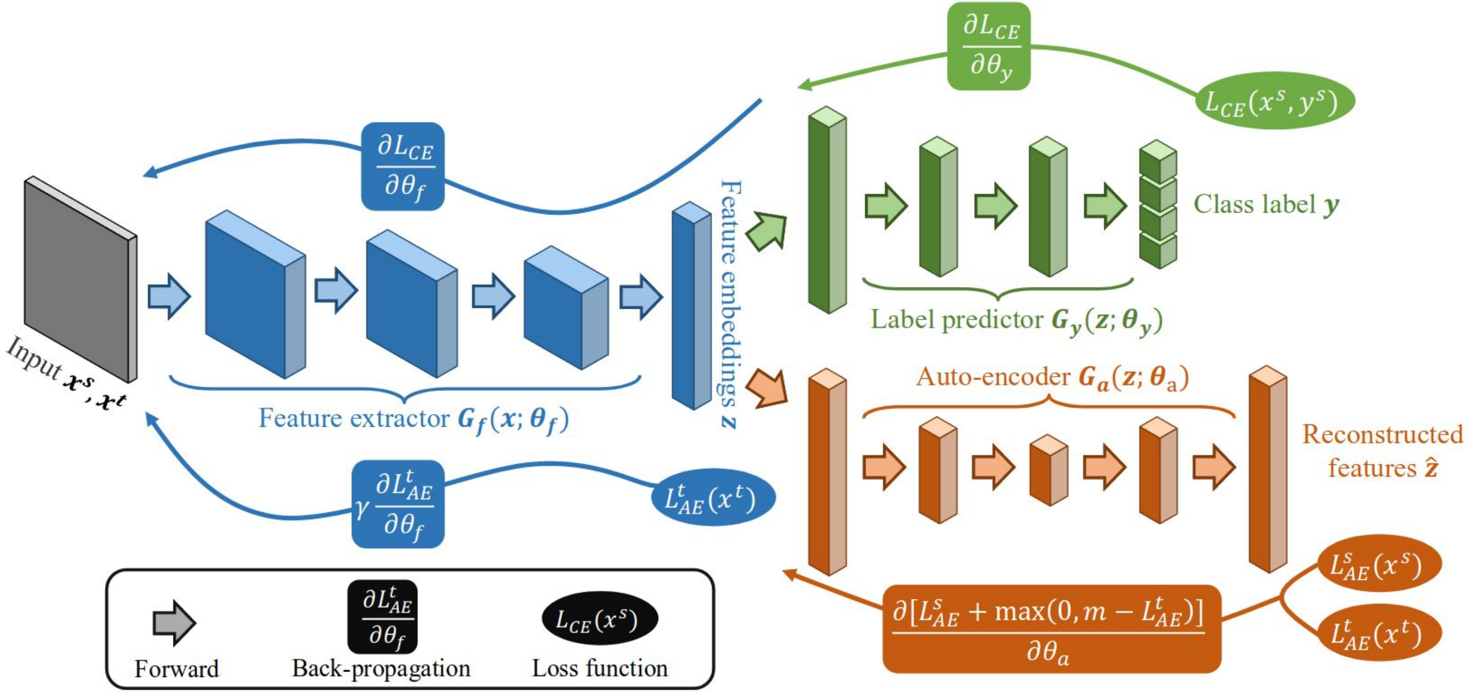
Ganin, Y., et al. “Domain-adversarial training of neural networks”. The Journal of Machine Learning Research, 17(1), 2096-2030, 2016.

Adversarial Domain Adaptation (RevGrad-DANN)



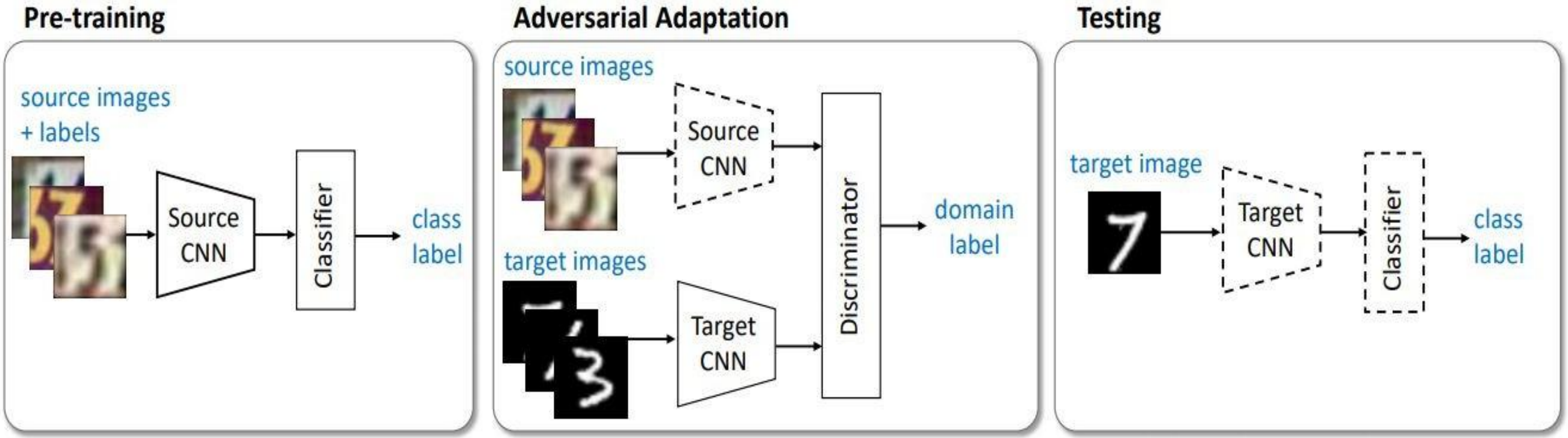
Ganin, Y., et al. “Domain-adversarial training of neural networks”. The Journal of Machine Learning Research, 17(1), 2096-2030, 2016.

Asymmetric Adversarial Domain Adaptation



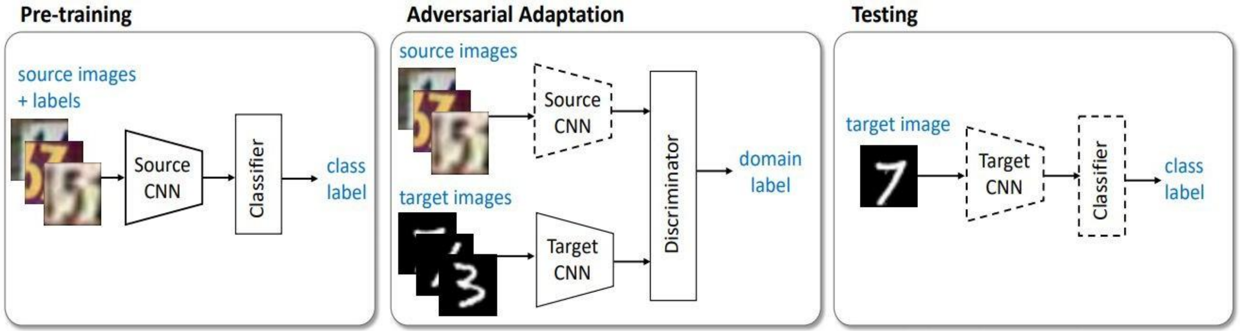
https://www.ecva.net/papers/eccv_2020/papers_ECCV/papers/123690579.pdf

Adversarial Domain Adaptation (ADDA)



Tzeng, E., Hoffman, J., Saenko, K., & Darrell, T., " Adversarial discriminative domain adaptation". In CVPR 2017.

Adversarial Domain Adaptation (ADDA)



$$\min_{M_s, C} \mathcal{L}_{\text{cls}}(\mathbf{X}_s, Y_s) = - \mathbb{E}_{(\mathbf{x}_s, y_s) \sim (\mathbf{X}_s, Y_s)} \sum_{k=1}^{K} \mathbb{1}_{[k=y_s]} \log C(M_s(\mathbf{x}_s))$$

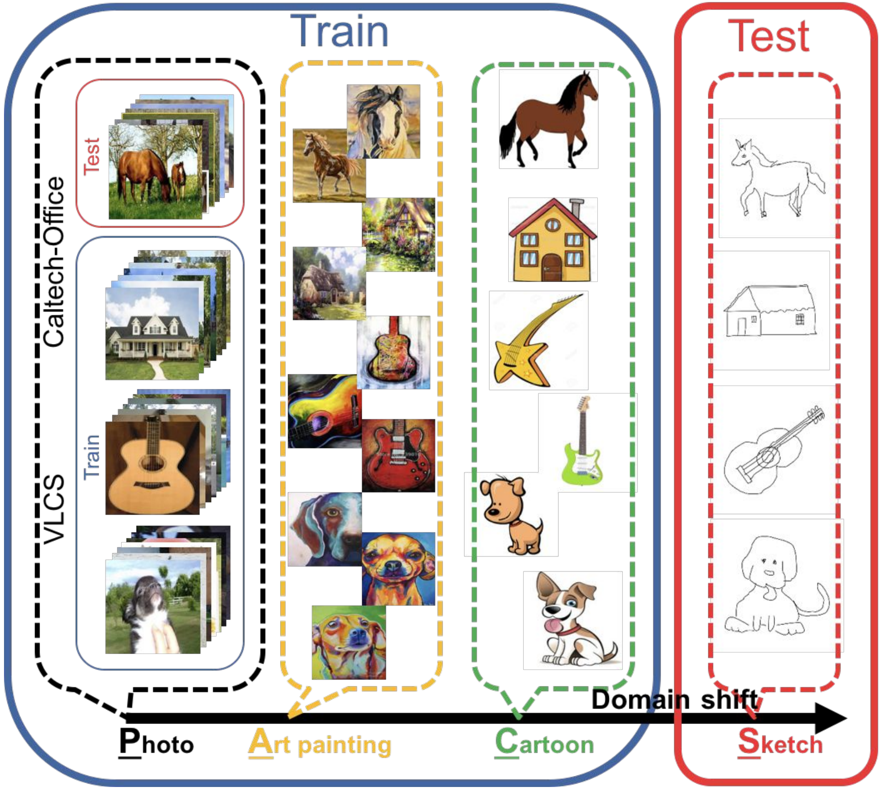
$$\min_{D} \mathcal{L}_{\text{adv}_D}(\mathbf{X}_s, \mathbf{X}_t, M_s, M_t) = - \mathbb{E}_{\mathbf{x}_s \sim \mathbf{X}_s} [\log D(M_s(\mathbf{x}_s))] - \mathbb{E}_{\mathbf{x}_t \sim \mathbf{X}_t} [\log(1 - D(M_t(\mathbf{x}_t)))]$$

$$\min_{M_t} \mathcal{L}_{\text{adv}_M}(\mathbf{X}_s, \mathbf{X}_t, D) = - \mathbb{E}_{\mathbf{x}_t \sim \mathbf{X}_t} [\log D(M_t(\mathbf{x}_t))].$$

Adversarial Domain Adaptation (ADDA)

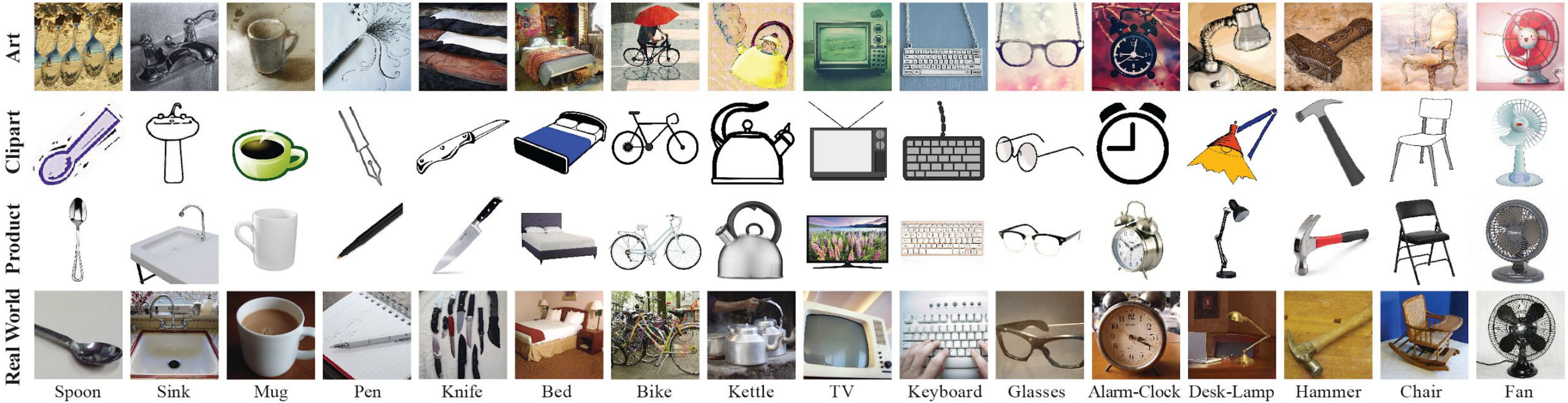
Method	$A \to W$	$D \to W$	$W \to D$
Source only (AlexNet)	0.642	0.961	0.978
Source only (ResNet-50)	0.626	0.961	0.986
DDC [5]	0.618	0.950	0.985
DAN [6]	0.685	0.960	0.990
DRCN [9]	0.687	0.964	0.990
DANN [19]	0.730	0.964	0.992
ADDA (Ours)	**0.751**	**0.970**	**0.996**

New Benchmark Datasets: PACS



http://www.eecs.qmul.ac.uk/~dl307/project_iccv2017

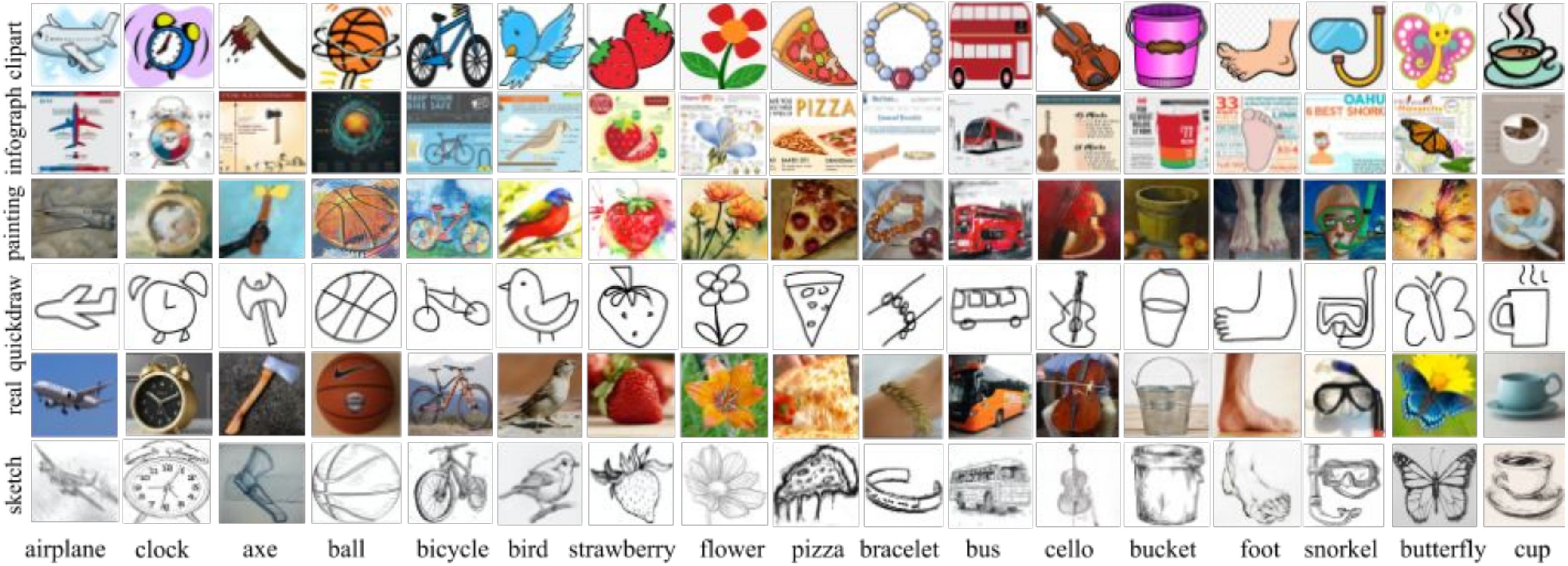
New Benchmark Datasets: Office-Home



Expt.	Ar→Cl	Ar→Pr	Ar→Rw	Cl→Ar	Cl→Pr	Cl→Rw	Pr→Ar	Pr→Cl	Pr→Rw	Rw→Ar	Rw→Cl	Rw→Pr	Avg.
GFK	21.60	31.72	38.83	21.63	34.94	34.20	24.52	25.73	42.92	32.88	28.96	50.89	32.40
TCA	19.93	32.08	35.71	19.00	31.36	31.74	21.92	23.64	42.12	30.74	27.15	48.68	30.34
CORAL	27.10	36.16	44.32	26.08	40.03	40.33	27.77	30.54	50.61	38.48	36.36	57.11	37.91
JDA	25.34	35.98	42.94	24.52	40.19	40.90	25.96	32.72	49.25	35.10	35.35	55.35	36.97
DAN	30.66	42.17	54.13	32.83	47.59	49.78	29.07	34.05	56.70	43.58	38.25	62.73	43.46
DANN	33.33	42.96	54.42	32.26	49.13	49.76	30.49	38.14	56.76	44.71	42.66	64.65	44.94
DAH-e	29.23	35.71	48.29	33.79	48.23	47.49	29.87	38.76	55.63	41.16	44.99	59.07	42.69
DAH	31.64	40.75	51.73	34.69	51.93	52.79	29.91	39.63	60.71	44.99	45.13	62.54	45.54

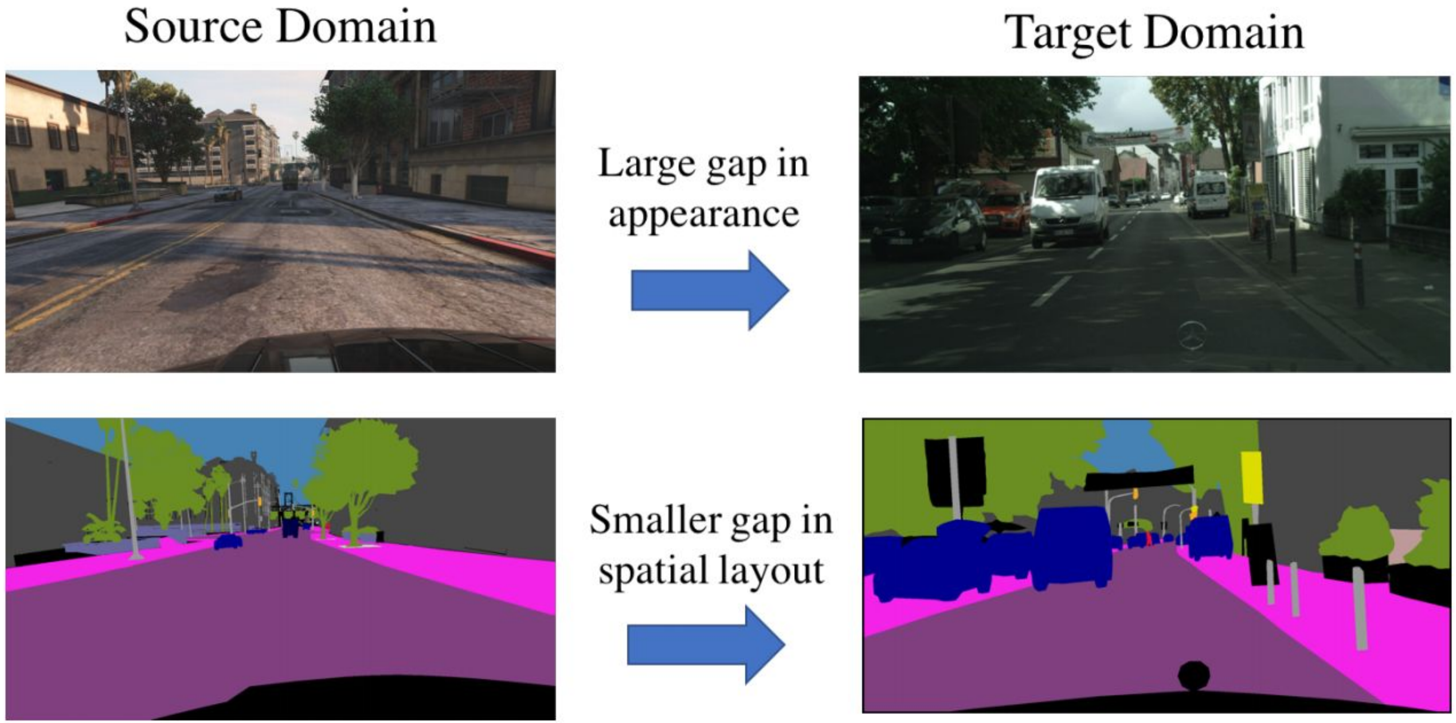
http://hemanthdv.org/OfficeHome-Dataset/

New Benchmark Datasets: DomainNet



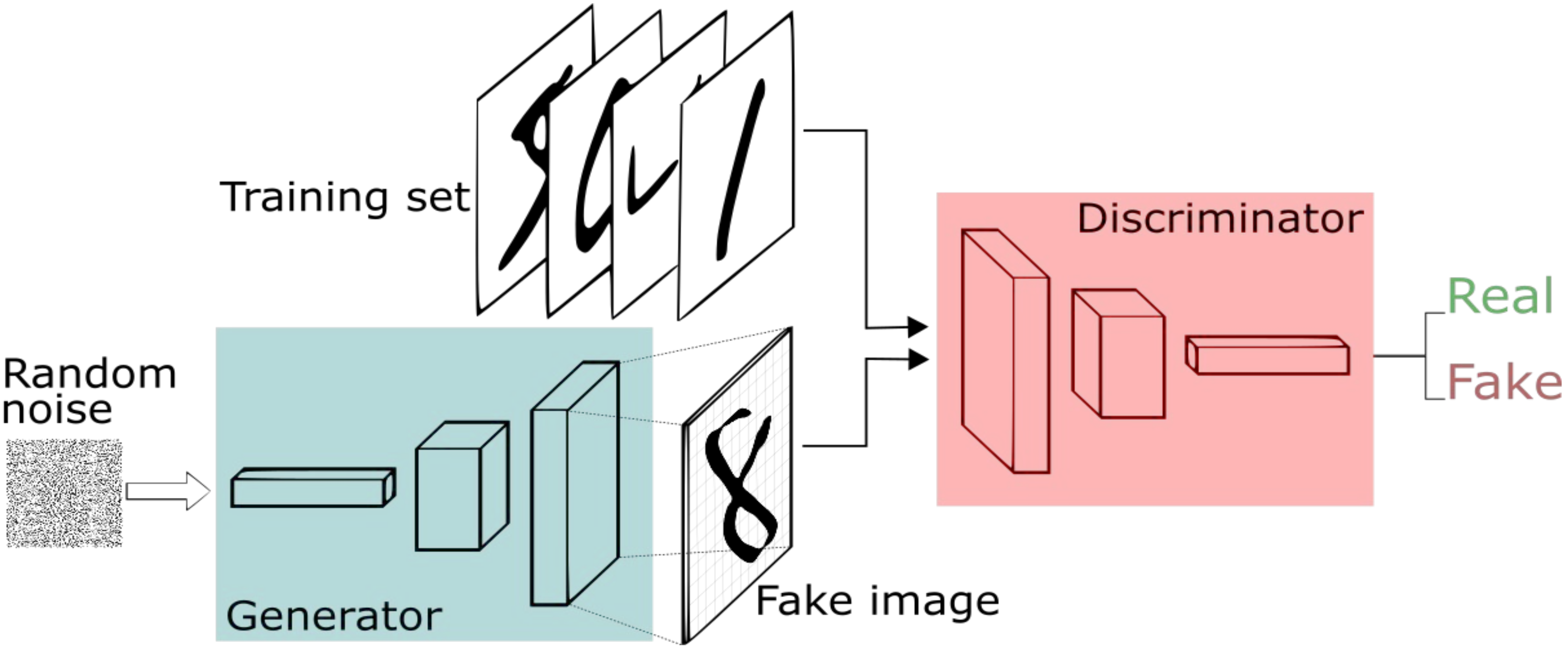
https://github.com/VisionLearningGroup/visda-2019-public/tree/master/multisource

New Benchmark Datasets: Semantic Segmentation



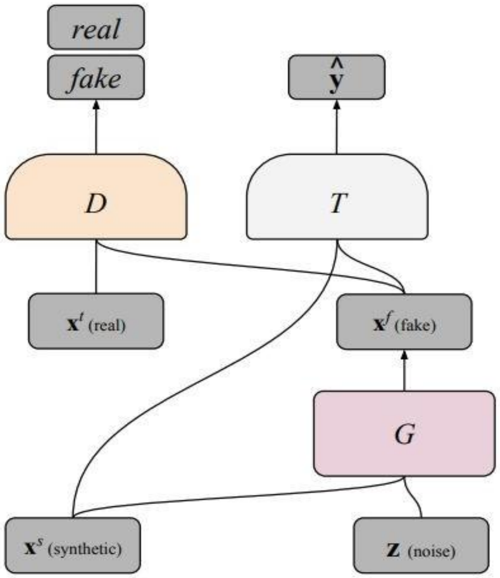
Tsai, Y. H. et al. “Learning to adapt structured output space for semantic segmentation”. In CVPR 2018.

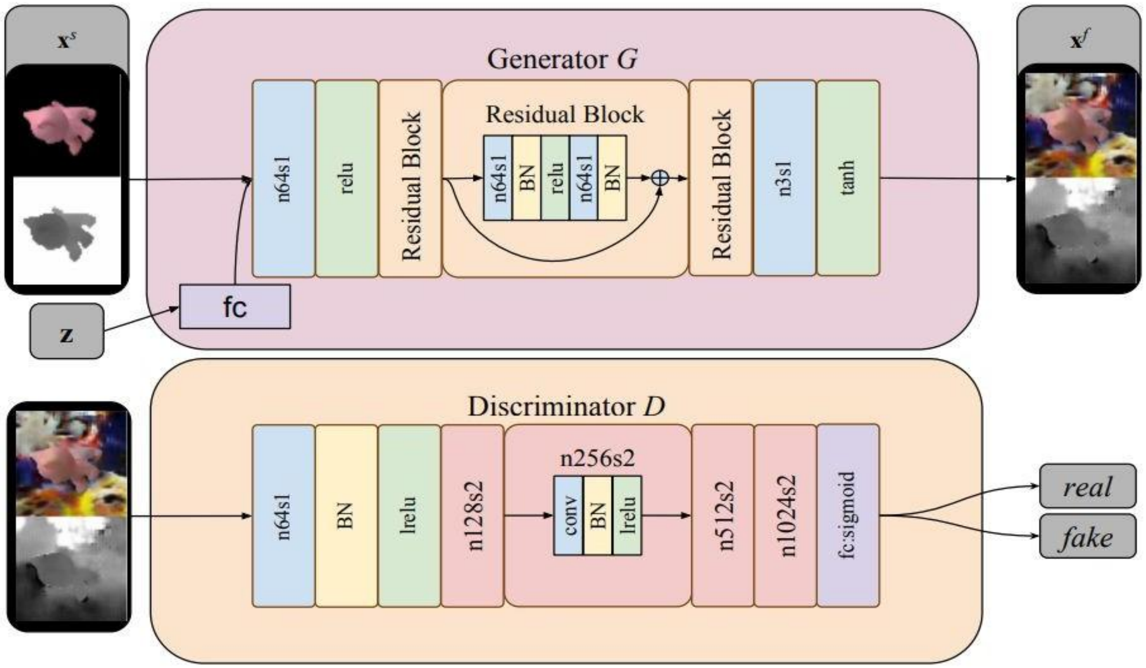
Pixel-Level DA (Generative Approaches)



https://skymind.ai/wiki/generative-adversarial-network-gan

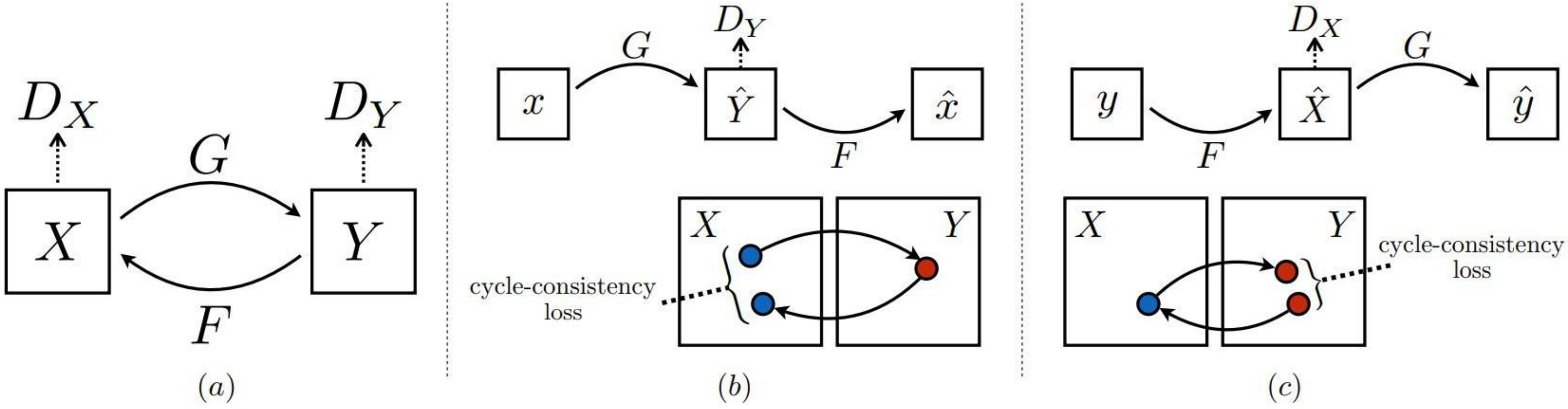
Pixel-Level DA (Generative Approaches)





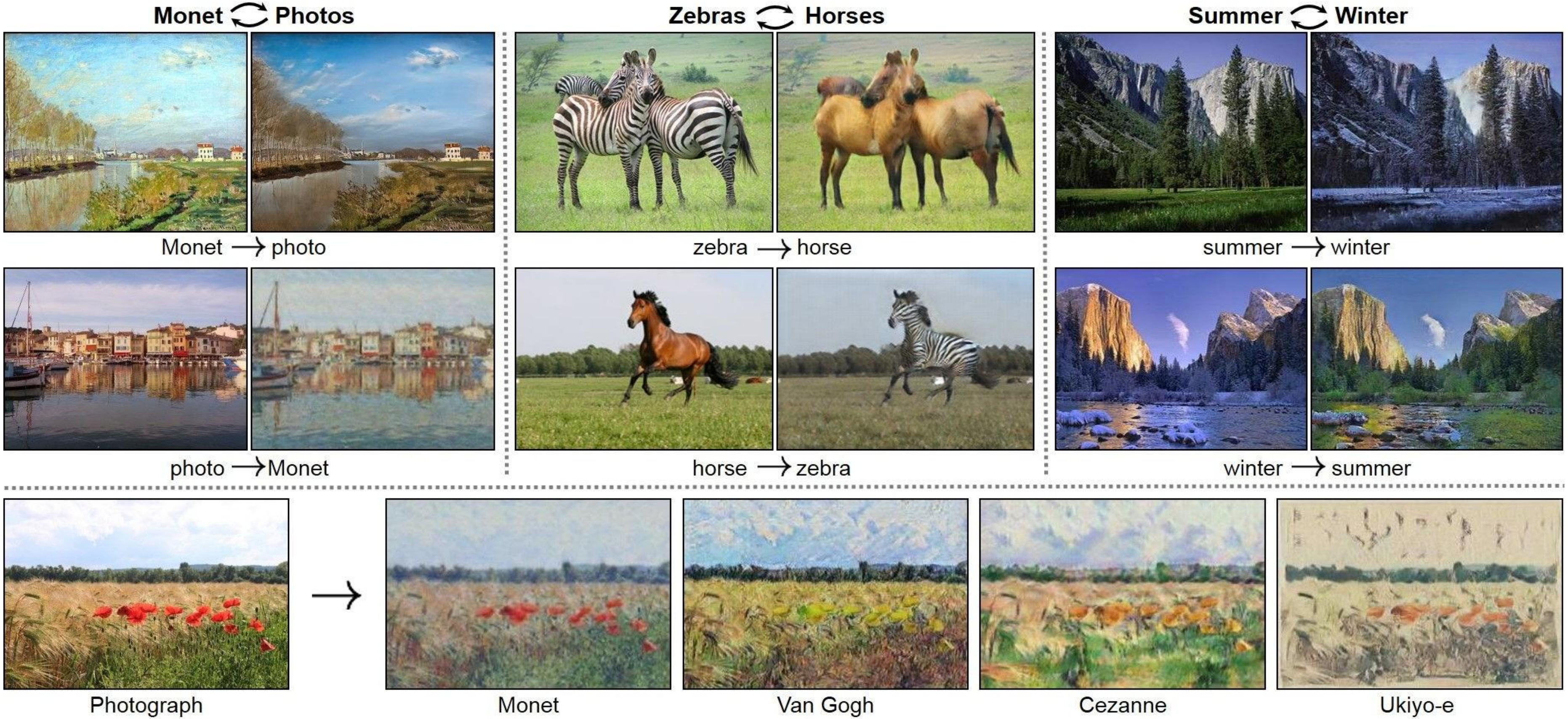
Bousmalis, K., Silberman, N., Dohan, D., Erhan, D., & Krishnan, D., “Unsupervised pixel-level domain adaptation with generative adversarial networks”. In CVPR 2017.

Unpaired Image-to-Image Translation: Cycle-GAN



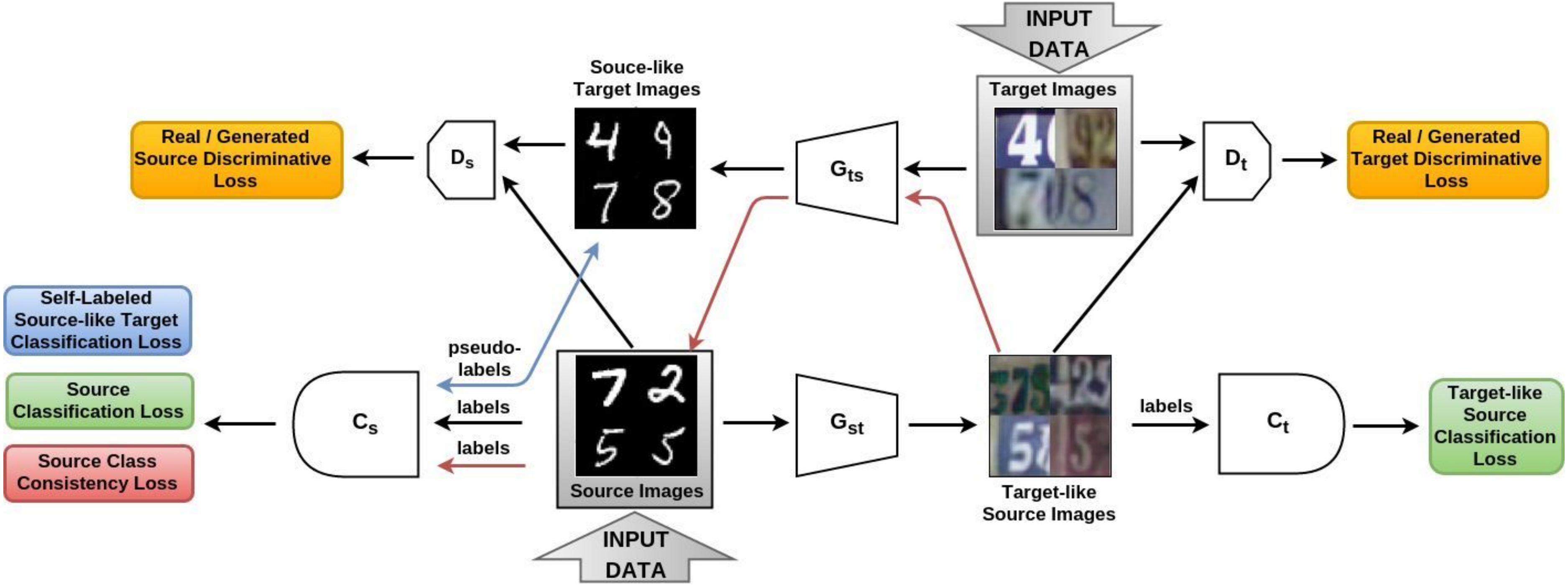
Zhu, J. Y., Park, T., Isola, P., & Efros, A.. “Unpaired image-to-image translation using cycle-consistent adversarial networks”. In ICCV 2017.

Cycle-GAN



Zhu, J. Y., Park, T., Isola, P., & Efros, A.. “Unpaired image-to-image translation using cycle-consistent adversarial networks”. In ICCV 2017.

SBADA-GAN



Russo, P., Carlucci, F. M., Tommasi, T., & Caputo, B. “From source to target and back: symmetric bi-directional adaptive gan”. In CVPR 2018.

SBADA-GAN

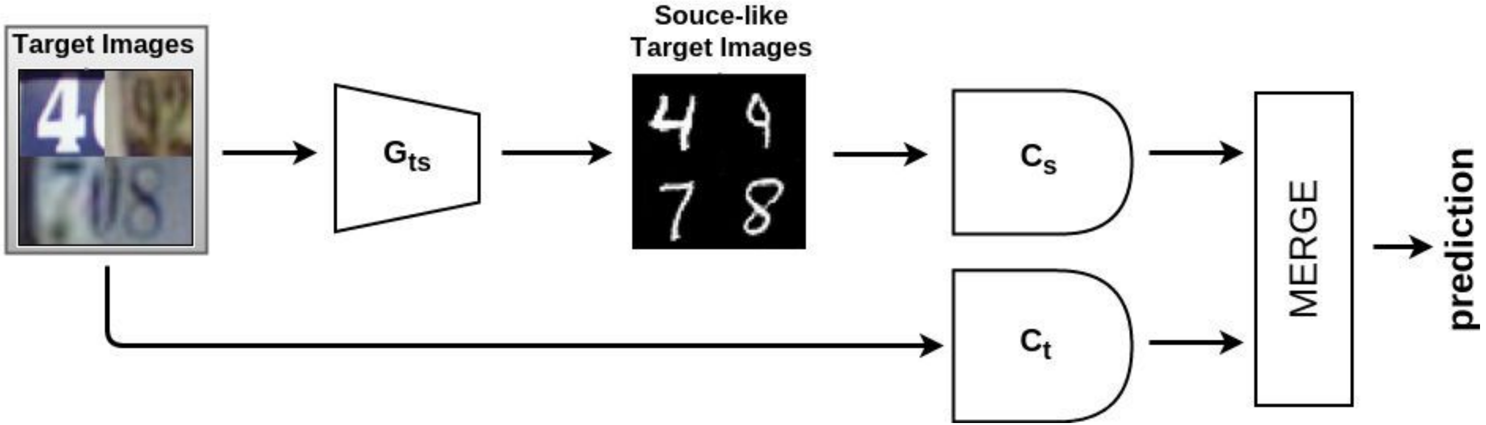
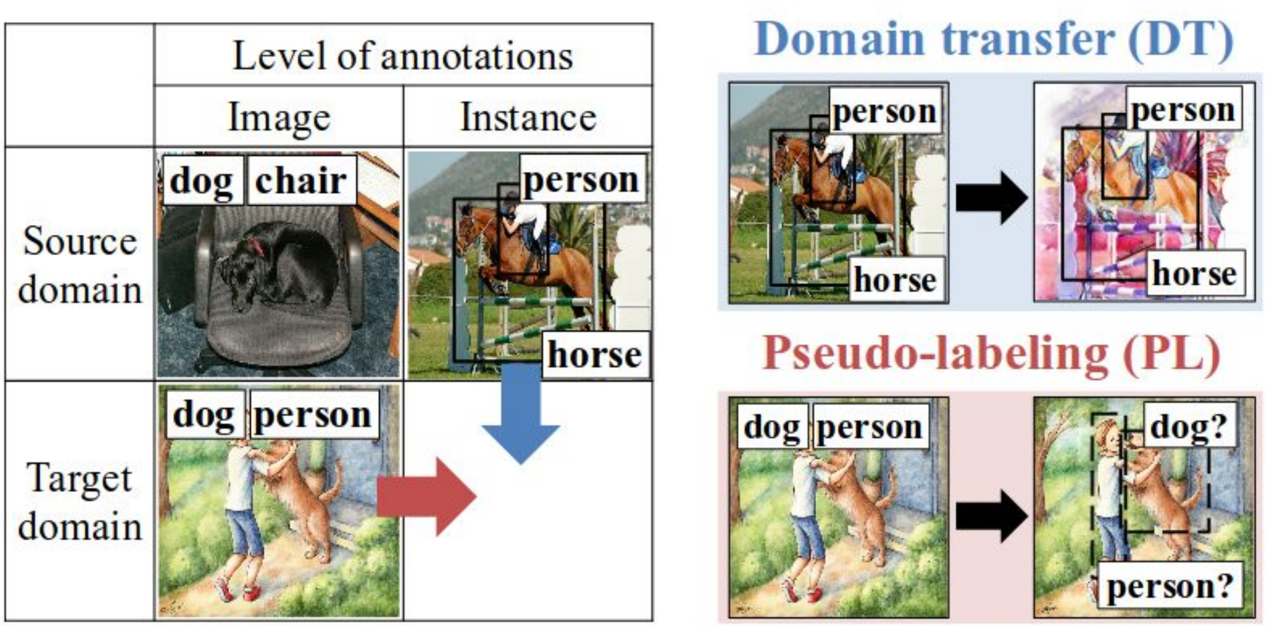


Figure 5: G_{ts} outputs (lower line) and their respective inputs (upper line) obtained with: (a) no consistency loss, (b) image-based cycle consistency loss [41, 17], (c) our class consistency loss. In (d) we show some real SVHN samples as a reference.

Pixel-Level DA in Detection



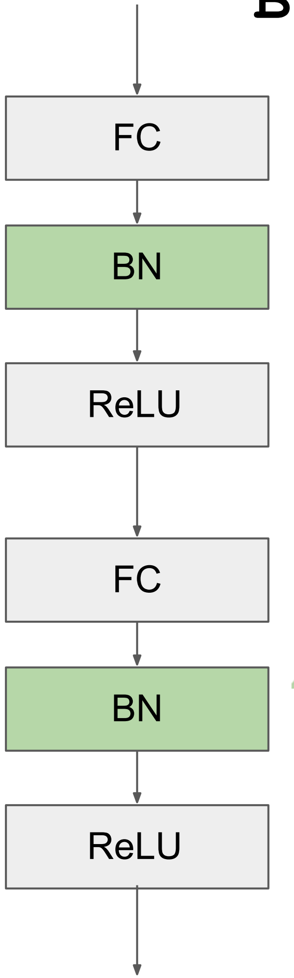
http://openaccess.thecvf.com/content_cvpr_2018/papers/Inoue_Cross-Domain_Weakly-Supervised_Object_CVPR_2018_paper.pdf

Pixel-Level DA in Detection

(a) Given image (b) Images with appearance shift

http://openaccess.thecvf.com/content_CVPR_2019/papers/Kim_Diversify_and_Match_A_Domain_Adaptive_Representation_Learning_Paradigm_for_CVPR_2019_paper.pdf

Batch-Normalization for DA



Input: Values of x over a mini-batch: $\mathcal{B} = \{x_{1\dots m}\}$;
Parameters to be learned: γ, β

Output: $\{y_i = \mathrm{BN}_{\gamma,\beta}(x_i)\}$

$$\mu_{\mathcal{B}} \leftarrow \frac{1}{m}\sum_{i=1}^{m} x_i \qquad \text{// mini-batch mean}$$

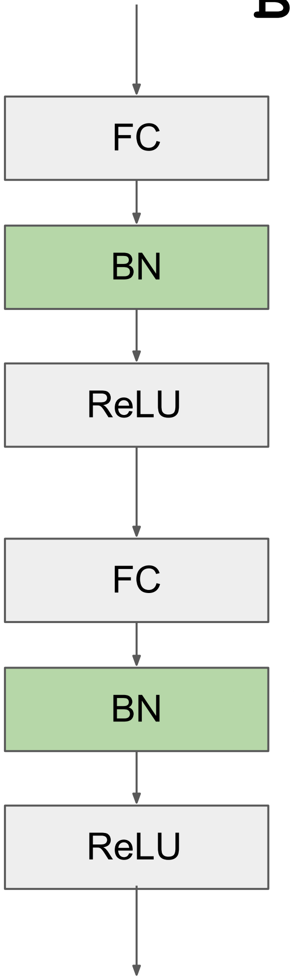
$$\sigma_{\mathcal{B}}^2 \leftarrow \frac{1}{m}\sum_{i=1}^{m} (x_i - \mu_{\mathcal{B}})^2 \qquad \text{// mini-batch variance}$$

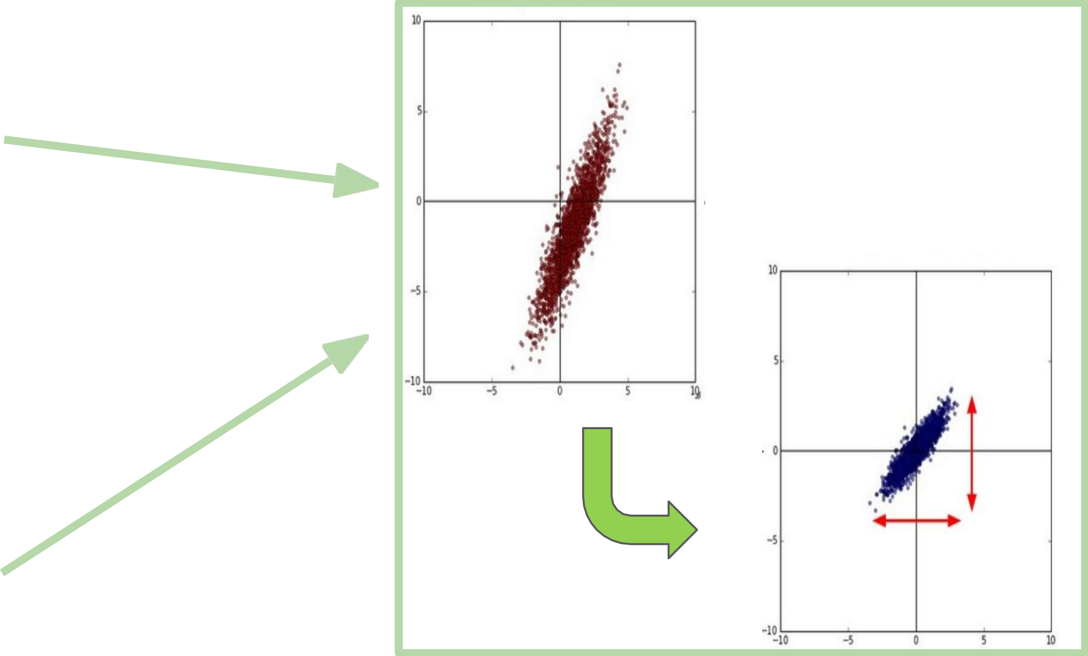
$$\widehat{x}_i \leftarrow \frac{x_i - \mu_{\mathcal{B}}}{\sqrt{\sigma_{\mathcal{B}}^2 + \epsilon}} \qquad \text{// normalize}$$

$$y_i \leftarrow \gamma \widehat{x}_i + \beta \equiv \mathrm{BN}_{\gamma,\beta}(x_i) \qquad \text{// scale and shift}$$

Ioffe, S., & Szegedy, C. “Batch Normalization: Accelerating Deep Network Training by Reducing Internal Covariate Shift: Automatic Domain Alignment Layers”. *ICML 2015*.
http://cs231n.github.io/neural-networks-2/#batchnorm

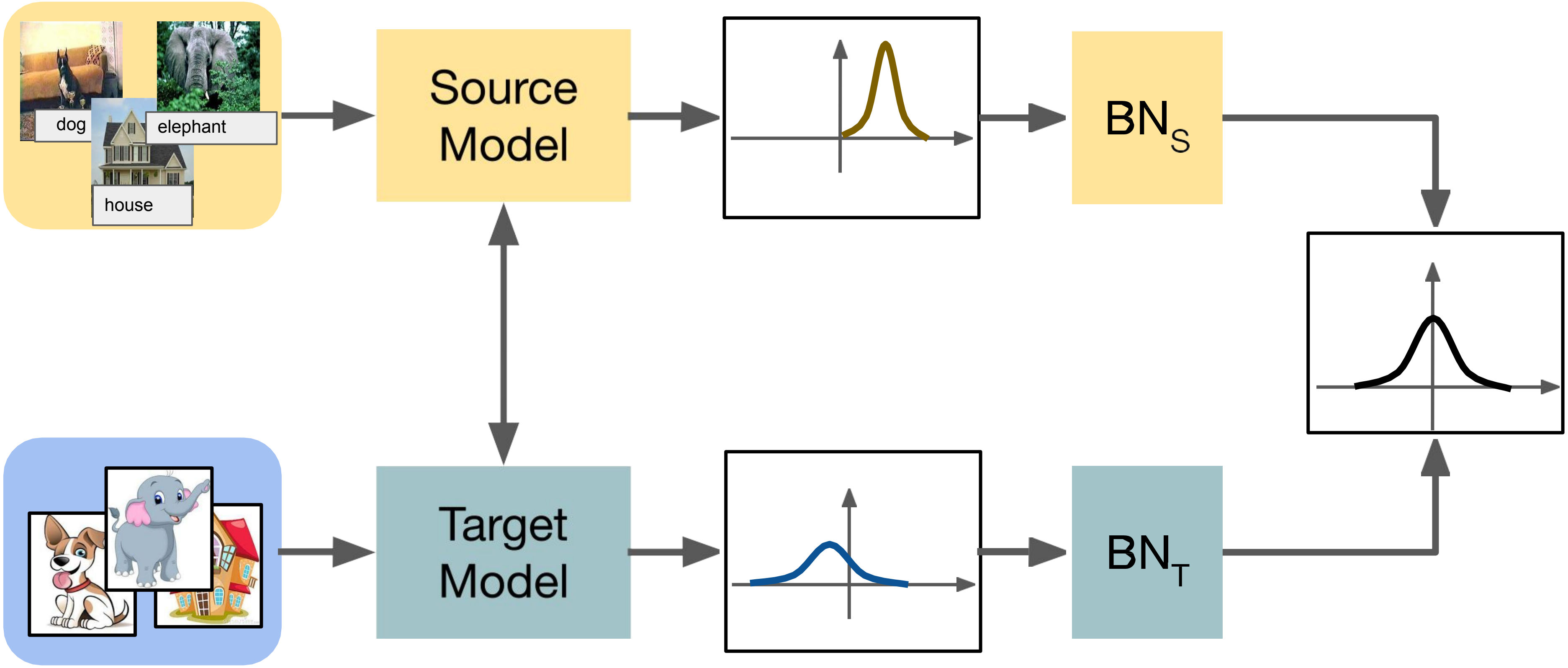
Batch-Normalization for DA





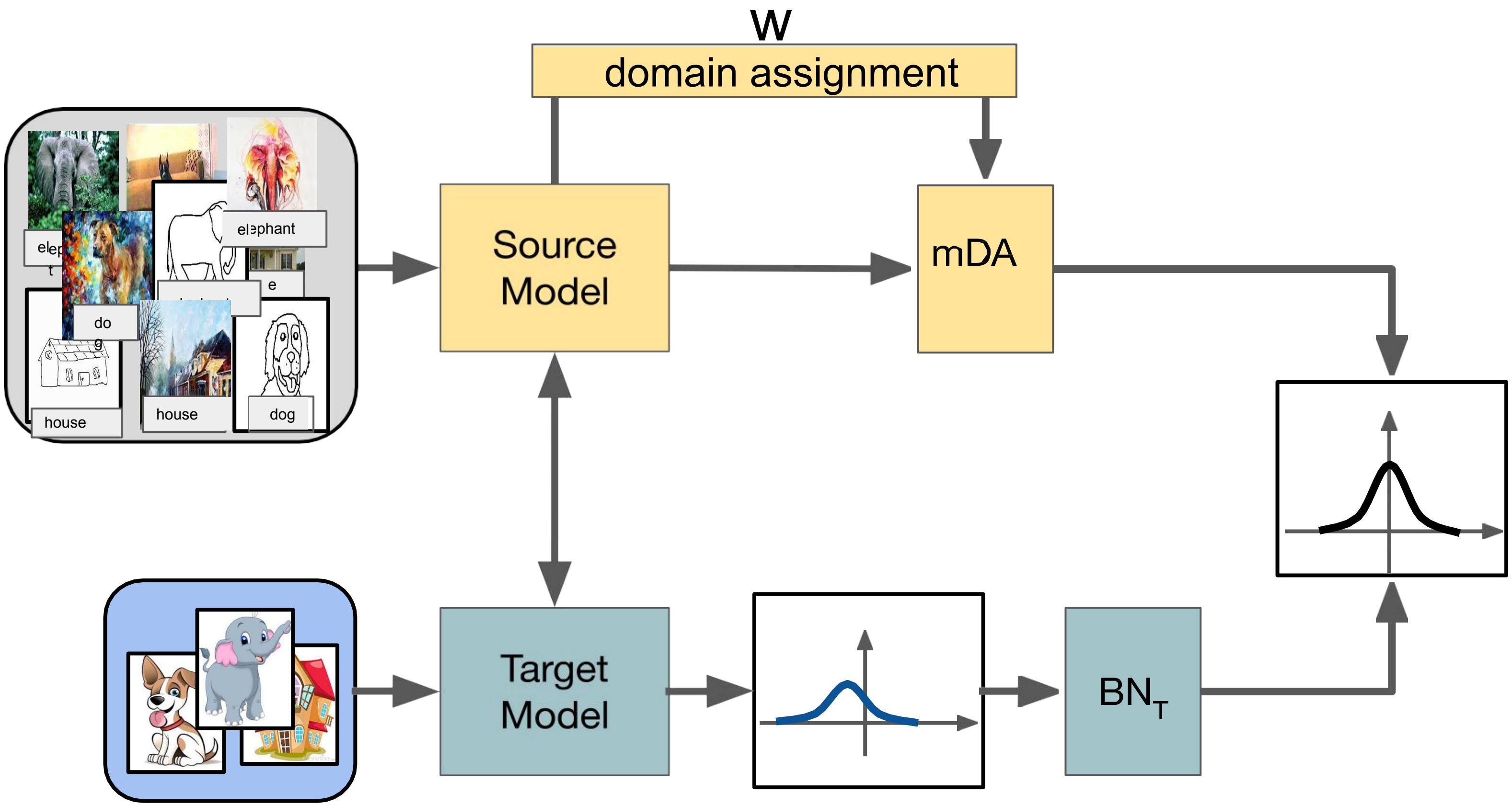
Ioffe, S., & Szegedy, C. “Batch Normalization: Accelerating Deep Network Training by Reducing Internal Covariate Shift: Automatic Domain Alignment Layers”. *ICML 2015*.
http://cs231n.github.io/neural-networks-2/#batchnorm

Batch-Normalization for DA



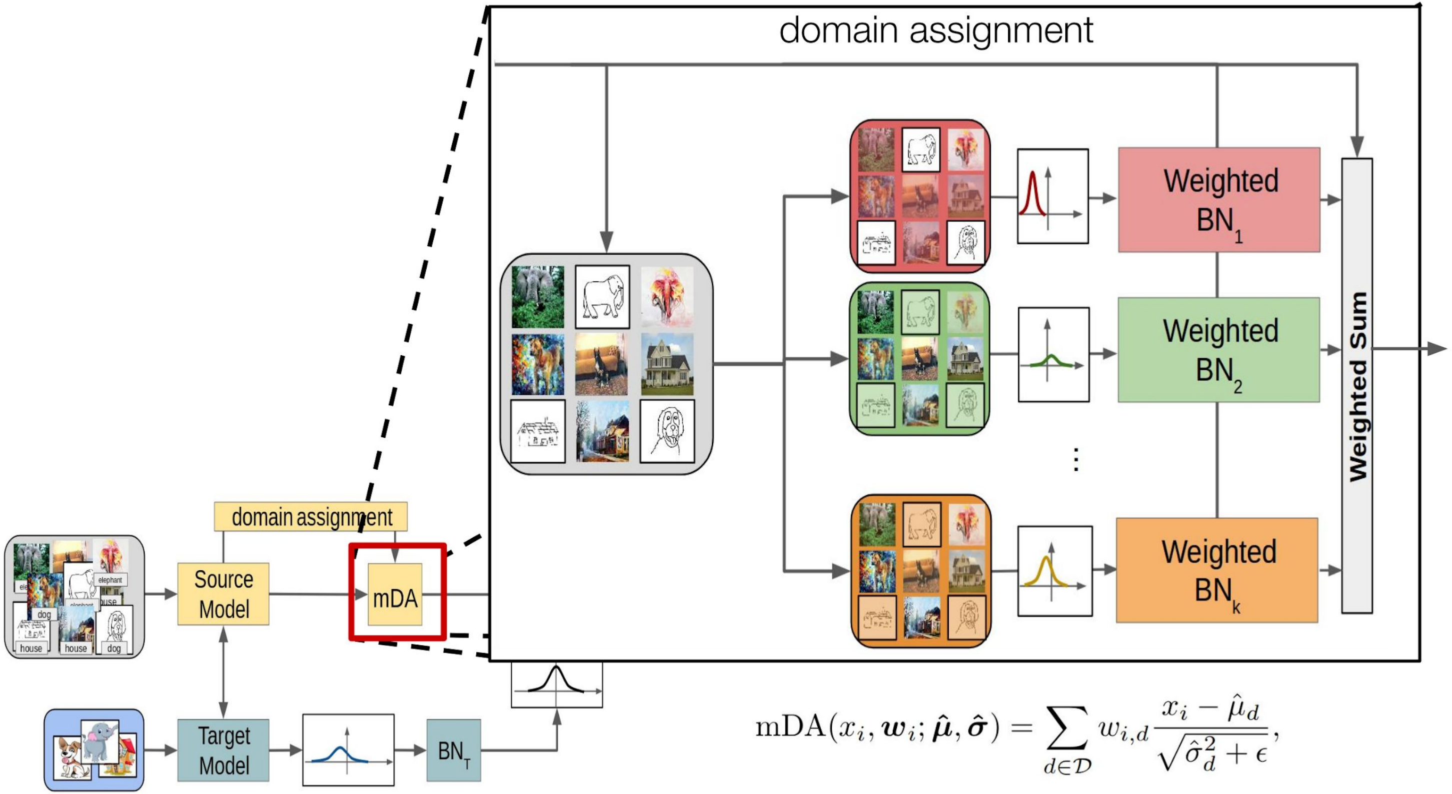
Carlucci, F. M., Porzi, L., Caputo, B., Ricci, E., & Rota Bulò, S. “AutoDIAL: Automatic Domain Alignment Layers”. *ICCV 2017*.
Carlucci, F. M., Porzi, L., Caputo, B., Ricci, E., & Rota Bulò, S. “Just dial: Domain alignment layers for unsupervised domain adaptation”. ICIAP 2017.

Multi-Source Batch-Normalization for DA



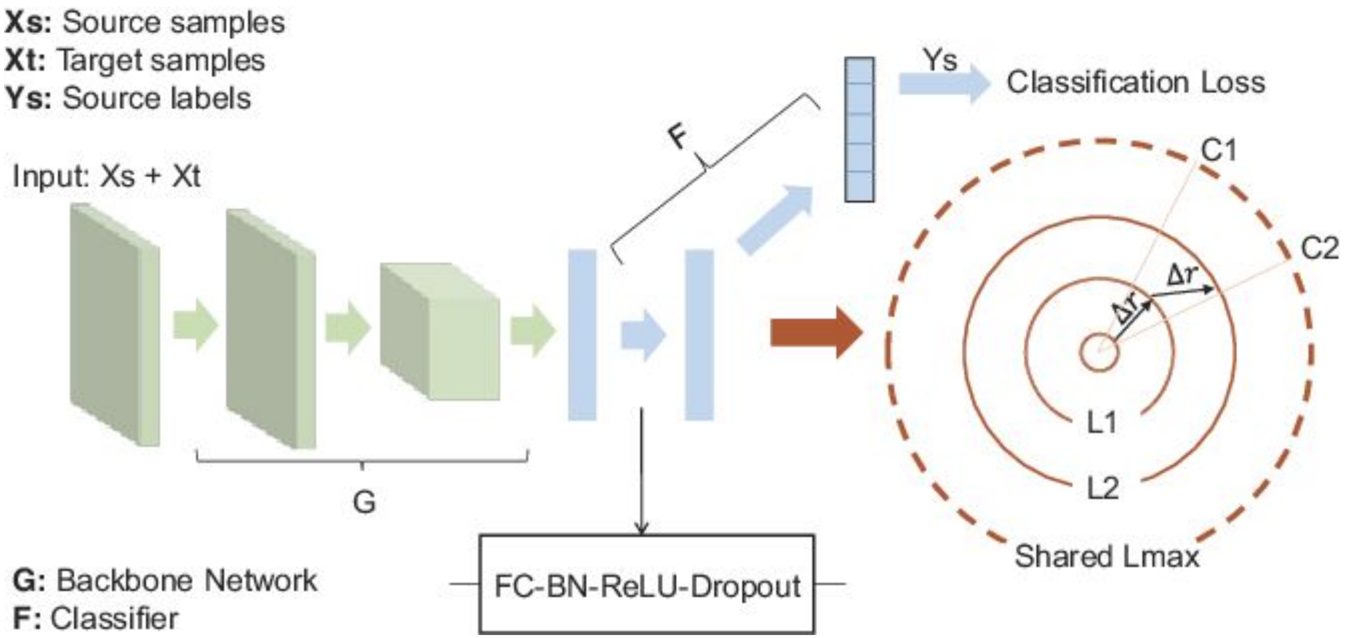
Mancini, M., Porzi, L., Rota Bulò, S., Caputo, B., & Ricci, E. “Boosting Domain Adaptation by Discovering Latent Domains”. *CVPR 2018.*

Multi-Source Batch-Normalization for DA



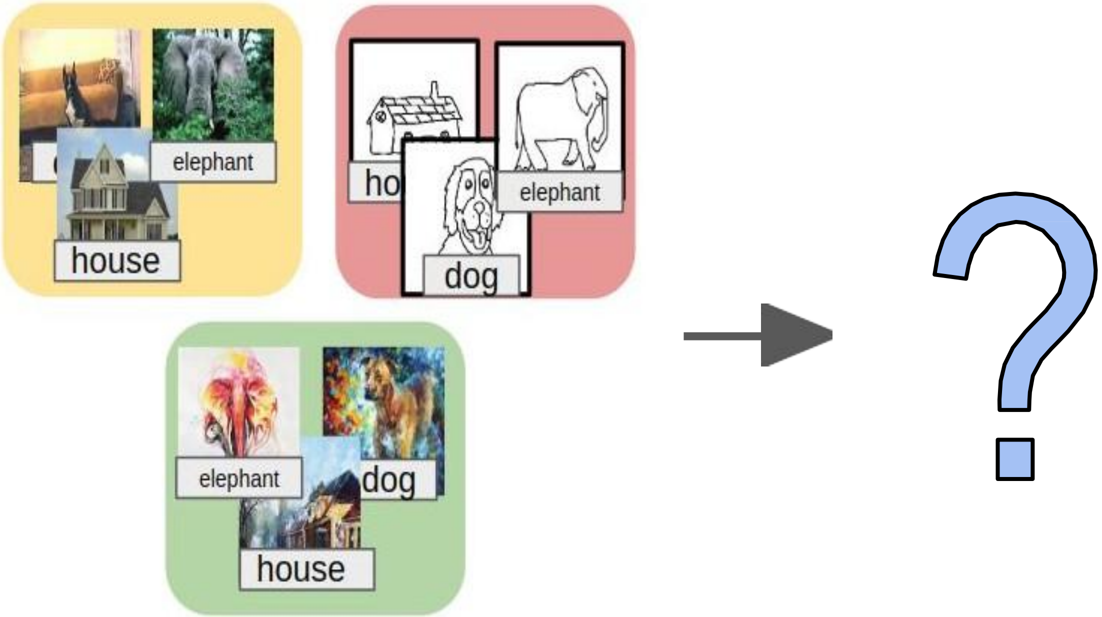
$$\mathrm{mDA}(x_i, \boldsymbol{w}_i; \hat{\boldsymbol{\mu}}, \hat{\boldsymbol{\sigma}}) = \sum_{d \in \mathcal{D}} w_{i,d} \frac{x_i - \hat{\mu}_d}{\sqrt{\hat{\sigma}_d^2 + \epsilon}},$$

Larger Norm More Transferrable

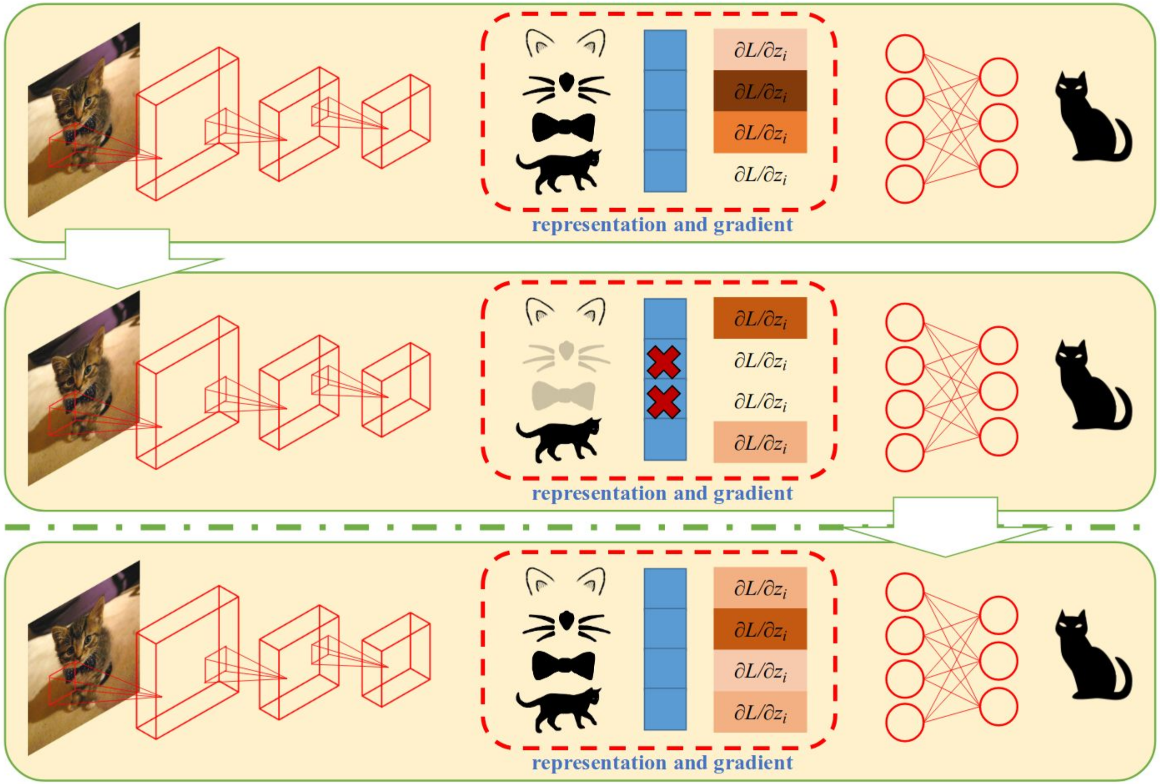


https://openaccess.thecvf.com/content_ICCV_2019/papers/Xu_Larger_Norm_More_Transferable_An_Adaptive_Feature_Norm_Approach_for_ICCV_2019_paper.pdf

Domain Generalization



Self-Challenging for Domain Generalization



https://www.ecva.net/papers/eccv_2020/papers_ECCV/papers/123470120.pdf